Michigan Business School

Case Book
2003 –2004

Michigan
Consulting
Club

MICHIGAN
BUSINESS SCHOOL

Table of Contents

About the Case BookProfitability Cases 4
Profitability Cases 5
Booz Allen & Hamilton - 1..17
Booz Allen & Hamilton - 2..18
Booz Allen & Hamilton - 3..21
Booz Allen & Hamilton - 4..23
Booz Allen & Hamilton - 5..24
Booz Allen & Hamilton - 5..26
Booz Allen & Hamilton - 6..32
Booz Allen & Hamilton - 7..36
Booz Allen & Hamilton - 8..37
Booz Allen & Hamilton - 8..39
Booz Allen & Hamilton - 9..42
Bain & Company- 1 45
Bain & Company- 2 47
Bain & Company- 3 48
Bain & Company- 4 50
Bain & Company- 5 51
Miscellaneous..52
Miscellaneous - 2..55
Cargill -1 57
McKinsey & Company - 1..60
McKinsey & Company - 2..62
McKinsey & Company - 3..63
McKinsey & Company - 4..64
McKinsey & Company - 5..66
McKinsey & Company - 6..68
McKinsey & Company - 6..70
AT Kearney- 1st Round - 1..74
AT Kearney 1st Round - 2..75
AT Kearney 1st Round- 3..77
AT Kearney 1st Round- 4..78
AT Kearney 1st Round- 5..81
AT Kearney 1st Round- 6..86
AT Kearney 1st Round- 7..89
AT Kearney 1st Round- 8..90
AT Kearney 1st Round- 9..93
AT Kearney (2nd Round) - 1 94
AT Kearney (2nd Round) - 2 103
BCG 1st Round- 1..107
BCG 1st Round- 2..108
BCG 1st Round- 3..111
BCG 1st Round- 5..117
BCG 1st Round- 6..119
BCG 1st Round- 7..120

Deleted: 5
Deleted: 6
Deleted: 9
Deleted: 11
Deleted: 12
Deleted: 14
Deleted: 20
Deleted: 24
Deleted: 25
Deleted: 27
Deleted: 30
Deleted: 33
Deleted: 35
Deleted: 37
Deleted: 39
Deleted: 40
Deleted: 41
Deleted: 44
Deleted: 47
Deleted: 50
Deleted: 52
Deleted: 53
Deleted: 54
Deleted: 56
Deleted: 58
Deleted: 60
Deleted: 64
Deleted: 65
Deleted: 67
Deleted: 68
Deleted: 71
Deleted: 76
Deleted: 79
Deleted: 80
Deleted: 83
Deleted: 84
Deleted: 93
Deleted: 97
Deleted: 98
Deleted: 101
Deleted: 107
Deleted: 109
Deleted: 110

BCG 2^{nd} Round- 1 122
BCG 2^{nd} Round- 2 125
BCG 2^{nd} Round- 3 127
BCG 2^{nd} Round- 6 129
DiamondCluster International – 2nd Round 131
DiamondCluster 2^{nd} Round- 1 134
DiamondCluster 1^{ST} Round- 1 136
DiamondCluster 1^{st} Round- 1 137

Deleted: 112
Deleted: 115
Deleted: 117
Deleted: 119
Deleted: 121
Deleted: 124
Deleted: 126
Deleted: 127

About the Case Book

Interview Education Guide

Introduction

This guide is meant to be a supporting document to the presentations and workshops held by the Michigan Consulting Club. It attempts to lay out some of the important issues that could arise when dealing with consulting case interviews. It is by no means an exhaustive guide to all the issues that could be important for cracking a case and should therefore be used as a starting point for thinking about cases, not an end in itself. In addition, since this is the first time such an education guide has been compiled, we would truly appreciate all comments and remarks regarding the guide as we work to improve it for future generations (We should set up a mechanism to collect those comments).

The cases

As you will see from this book, we have broken down consulting interview cases into 4 major types:

- Profitability
- Growth and market entry
- Operations
- Economics

You will also find brainteasers and market sizing problems that simply test your logical thinking skills and often do not require an extensive knowledge of business issues.

Deleted: some brain-teasers

Finally, it should be noted that many cases tend to combine these different types. Do not be surprised if in the real interview what started as a growth case becomes a profitability case. After all, this happens in the real world all the time: a consulting firm is brought in to dream up a new growth strategy for the ambitious new CEO only to realize that basic cost issues must be dealt with before focusing on growth.

Practicing cases

One key to practicing cases is that **you must focus on quality**, not quantity. Before cracking too many cases, you should think through the issues and frameworks that may be important for each type of case. The purpose of this Education Guide is to help you with that. However, is should be stressed that there is no one "right" approach to any case, so it is difficult to compile an exhaustive list of issues that may arise. Nevertheless, it is important that you think about the relationships between the issues. One good way

Formatted

of doing this is drawing "mind maps" of all the possible issues that are important for a specific problem type.

Deleted: The frameworks below are based on my mind maps that I created before interviewing with consulting firms in my first year.¶

Cracking the case

Some tips and tricks to consider when cracking cases:

Deleted: watch out for

- After listening to the case question, **paraphrase the problem** in a few short sentences. Confirm with the interviewer that you have understood all the main issues. Ask any clarifying questions if needed.
- Take a few moments (not more than 30 seconds - 1 minute depending on the length of the case) to lay out an **initial framework** that will help you approach this problem. **Be flexible in modifying your framework** as new information arises.
- **Don't force a framework**!!! By no means apply the frameworks described in this interview education guide directly. First of all, they are way too exhaustive, the cases you face will focus on a few of the possible range of issues. Second, and more importantly, each case is unique, so it needs a unique approach tailored to the case. Listen to all the details given in the case to help you tailor your approach.
- Your **structure** is more important than the end result: therefore, always concentrate on articulating your thinking process.
- At the end of the case **SYNTHESIZE** (not summarize!!!)[a] the key issues and make a few recommendations. Focus on the 'so what?', i.e. why are these issues that you are highlighting important to the firm and how will your recommendations impact the business going forward. Importantly, state the trade-offs if the firm decides to implement your recommendations.

[a] Synthesizing means stating the problem in one sentence, then highlighting the 2-3 most important issues you have uncovered and finally providing possible solutions to those issues. Summarizing means briefly restating all the main points of your discussion. This may not be easy to learn, but make sure to do the former and not the latter!

Chapter One: Profitability cases

If you are dealing with a case where the firm faces declining profits, the general framework to follow is:

Profitability

- **Costs**
  - **Fixed costs**
  - **Variable costs**
- **Revenue**
  - **Price**
  - **Quantity**
- **Mix**
  - **Product mix**
  - **Customer segment mix**

Before going deeper into what issues might arise in each of these buckets, a few general comments:

- When applying a framework such as the one above, make sure you pay attention to the order of the issues. Generally, don't start with an issue at random. If you need to, ask a few simple questions (e.g. what has happened to revenues in the past year years? If they have been decreasing, this may be a good area to start) to decide which bucket to do a deep dive in first.
- Don't apply this framework rigidly. For example, a project may not have any fixed costs or there may be a better classification of costs based on the initial information you are given.

Issue "mind-map" for profitability cases

This mind map attempt to synthesize the main issues and factors in each of the 'buckets' (costs, revenue and mix) of a profitability case. For more on the individual factors and issues, consult your class notes and the Consulting Job juice cards.

Deleted: an

Profitability

- **COSTS**

TOOLS FOR ANALYZING COSTS:
Competitive benchmarking
- **Analyze industry scale curve**
- **Relative cost structures of competitors**
- **Cost advantages vis-à-vis competitors**

  - Fixed costs
    - Capacity utilization
    - Economies of scale
    - Economies of scope

TOOLS FOR ANALYZING COSTS:
Break-even analysis
Break-even Q = FC / (P – VC)

  - Variable costs
    - Raw materials
    - Labor
    - Transportation
    - Overhead
    - Selling, general & administrative
    - Other distribution c
    - Depreciation
    - Financing costs

TOOLS FOR ANALYZING COSTS:
Sizing total costs:
- **Identify major cost buckets (Pareto principle)**
- **Identify changes over time**
- **Which ones can we impact?**

Process flow diagrams
- **Which costs are highest in the system?**

Deleted: Where

Deleted: are the

SAMPLE RECOMMENDATIONS CONCERNING COSTS:
- **Consolidate supplier network**
- **Modify incentive system to achieve cost control**
- **Outsourcing**
- **Consider horizontal/vertical integration to achieve cost synergies**
- **Implement new IT systems to better monitor costs**

WHAT ARE THE COSTS OF IMPLEMENTING THE PROPOSED COST SAVINGS?

- **REVENUES**
  - Price
    - Target segments

ANALYZING REVENUES:
- **Bargaining power**
- **Consider switching costs**

    - Price elasticity of demand, current demand

ANALYZING REVENUES:
- **Demand analysis**

- Substitute prices
- Perceived value-added →

ANALYZING REVENUES:
- **What is the customer's "willingness-to-pay"?**

- Differentiation →

ANALYZING REVENUES:
- **Bases of differentiation:**
  - **Quality, design, location, service or simply price?**

- Competitive pricing →

ANALYZING REVENUES:
- **Game theory**
- **Price retaliation**

SAMPLE RECOMMENDATIONS CONCERNING PRICING:
- **Skimming vs. penetration pricing strategy**
- **Premium vs. value product**
- **Niche vs. market product**
- **Price discrimination**
  - **Bundling**
  - **Warranties**
  - **Two-part tariffs**

- Quantity
  - Market share analysis
  - Impact of new entrants
  - Growth patterns in demand
  - Capacity utilization in existing/planned facilities
  - Types of customers gained or lost
  - Seasonality, cyclicality of demand
  - Demand for substitutes and complements
- **MIX**
  - Product mix
  - Customer segment mix

ANALYZING PRODUCT PORTFOLIOS:
- **BCG growth-share matrix**
- **GE-McKinsey planning grid**
- **Optimal resource allocation (e.g. financial analysis using NPV)**

CONSIDER IMPACT OF STRATEGY:
- **Cannibalization of other products**
- **Impact on brand equity, reputation**

Chapter Two: Growth and market entry cases

These are perhaps the most common and the most complex cases, because the issues are numerous and inter-related. In general, one quick-and-easy framework that might be applied to many growth cases is the 3 C model:

Growth problem = issues related to:
- **Company**
- **Competition**
- **Customers**

Issue "mind-map" for market entry cases

Many growth cases require you to analyze a new market entry or new product opportunity. The below "mind-map" will help you get an overview of all the issues potentially involved. Some growth cases will not be about evaluating a new market entry opportunity. In this case, the framework will need to be modified, but the issues that may come up will tend to be those discussed below.

Evaluating market entry

1. **GO or NO GO?** → **Cost-benefit analysis**
    - **Target market**
        - **External environment** (watch out for international cases: if the case states that this is outside the US, chances are there will be specific issues you will have to consider that arise from that environment)
            - Regulatory
                - As a barrier to entry (e.g. domestic content restrictions in procurement, public authority controls, like licenses and approved suppliers and trade barriers, like quotas and tariffs)
                - As price influencer (e.g. direct or indirect subsidiaries)
                - As competitive force (e.g. patent, intellectual property, anti-trust laws)
            - Technology
            - Economy
                - GDP cycle
                - Inflation
                - Exchange rate
            - Social factors
                - Demographics

- Size and growth of market
- Company
    - Rationale for diversification
        - Invest excess cash flow
        - Increase market share (due to economies of scope or positive impact on core business)
        - Increased grow th
    - Resources available
    - Core competences

Opportunity for differentiation?
- **Premium pricing opp'y (skimming vs. penetration pricing strategy)**

    - Cost structure
        - Cost drivers
        - Economies of scale
        - Outsourcing production? (may be an important consideration, because the company might have no knowledge of producing products for this new market segment)
    - Capacity issues
        - Minimum efficient scale
        - Utilization, excess capacity
    - Product mix
    - Management capabilities
- Competitors
    - Structure of industry
        - Number of players
        - Vertical/horizontal integration linkages → **Value chain**
        - Market shares
        - Capacity issues (presence of excess capacity)
        - Strategic positioning
        - Financial performance
    - Basis of competition
        - Cost-based (price wars)
        - Technology-based (innovation)
        - Cooperation (duopoly)
    - Retaliation
    - Value proposition vis-à-vis competitors

- Suppliers
  - What are our key needs:
    - Low cost
    - High quality (quality control, low defect rates)
    - Timeliness (JIT, quick turnaround)
    - Service
  - Bargaining power
    - Depends on concentration of suppliers, product differentiation (switching costs and substitutes), possibility of forward integration
- Customers
  - New and existing customer segments
    - Profitability (revenue depends on trial and repeat rate for a new product)
    - Customer needs (commodity vs. differentiated products)
    - Bargaining power (higher with few large customers, standardized product which constitutes a large part of the total cost, customers with low margins and threat of backward integration)
- Financing
  - Internal financing
  - Debt
  - Equity

FINANCIAL ANALYSIS TOOLS:
1. **NPV analysis (CF, r)**
2. **ROCE (capital, profits, hurdle rate)**
3. **Payback, IRR, profitability index**
4. **Break-even analysis**

- **Ability to build competitive advantage**
  - Company (see above)
    - Ability to transfer core resources, competences
  - Current competitors (see above)
  - Threat of new entrants
    - Economies of scale
    - Learning curve
    - Access to distribution
    - Product differentiation
    - Government barriers
    - Capital requirements
- **Sustainability of strategy**
  - Future market growth
  - Number of new market entrants expected

- **Linkages between existing/new businesses**
  - Synergies
  - Competitive tactics (e.g. positive impact on core business)
  - Cannibalization threat (market cross-elasticities)
  - Gain access to valuable distribution channels

2. **METHOD OF ENTRY**
   - **Mergers and acquisitions**
     - Post-merger integration (PMI) issues:
       - Implications for firm culture
       - Strategic fit
       - Exit strategies
     - Compare margins of acquirer/acquiree
     - Synergies
     - Dilution effect on EPS
     - Tax, legal issues
     - Anti-trust issues
     - Competitive retaliation (consolidation "waves")
   - **Alliances (joint ventures and others)**
     - Advantages
       - Exploits key internal resources
       - Quicker access to market or technology
       - Lower cost, risk
       - Shared management responsibility
     - Disadvantages
       - Need to agree of partner contribution, labor practices, quality
       - Firms may be competitors (strategic intent?)
       - Lack of control?
       - Differences in culture, management styles
       - Need to capture, appropriate other's skills to succeed
   - **Internal growth**
     - Exporting
     - Geographic expansion (FDI, store opening)
     - Licensing
     - Private labeling

3. **ALTERNATIVE TYPES OF DIVERSIFICATION**
   - **Geographic**
     - Domestic
     - International

- **Vertical integration**
  - Supplier bargaining power becomes a non-issue
  - Smoother production flow (?)
  - Better access to markets
  - Distribution cost advantage
- **Related markets**
- **Unrelated markets**

Chapter Three: Operations cases

These cases are encountered less frequently, but some firms are more likely to have them than others. The "pure" strategy firms (BCG, McKinsey and Bain are less likely to have them, while those focusing more on implementation give them more often) As the operations core is taught in Winter A, it is a good idea to study the key concepts ahead of time, especially if you have made it to second rounds with a firm that does more operations-type work. **Please consult Prof. Anupindi's course review on the MCC site**.

Operational performance problem
= issues relating to:

- Process costs
- Capacity utilization
- Inventory
    - Reasons for inventory
        - Variability in demand
        - Economies of scale
        - Seasonality
    - Costs of inventory
        - Opportunity costs of capital
        - Warehousing costs (storage and handling)
        - Insurance costs
        - Obsolescence
        - Setup costs
- Customer needs
    - Importance of timely delivery
    - Quality
    - Demand trends, dynamics

Costs

Utilization

Inventory

Flow time

Quality

Flow rate, Throughput

Tips & tricks

- Use **process flow diagrams** when appropriate to help lay out the problem
- Focus on improvements to the **bottleneck resource**
- Take into account the **variability** in the system
    - Optimal utilization rate (customer satisfaction vs. efficient operations)
    - Safety stock
    - Bull-whip effects in supply chains
- **Little's Law** (I = R x T) is a powerful tool

Chapter Four: Economics cases

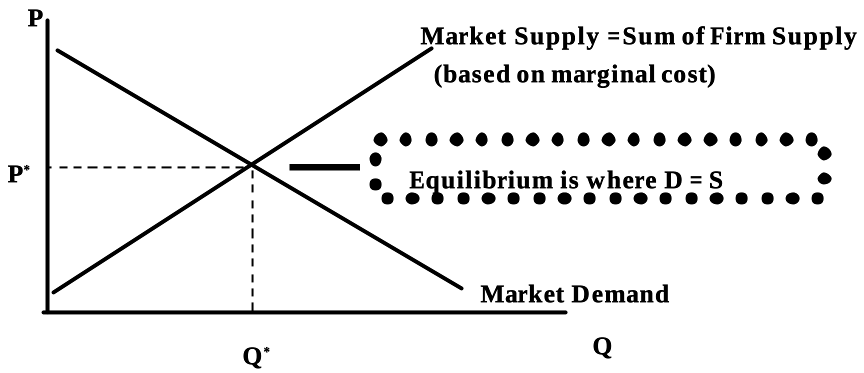
Often, Economics "cases" are embedded in profitability or growth cases. Typical tasks will involve making recommendations on **pricing**:

- What price the market will bear? Is the current price too low or too high?
- How will prices change in this industry in the future?
- How should we react to competitor price changes, how will they retaliate?
- How to enhance or support price discrimination?

The other commonly encountered economic issue is **production scale**. Some cases may call for estimating the Minimum Efficient Scale or dealing with economies of scale.

For a sample framework on how to approach price-related issues, see Chapter One: Profitability cases. For more on the economics concepts you will need for case interviews, see Prof. Suslow's course review on the MCC website.

The single most important tool for dealing with economic cases will be supply-demand analysis (particularly in competitive markets):



However, be sure to also review:

- Competitive markets vs. oligopolies vs. monopolies
- Elasticities
- Cost fundamentals
- Economies of scope and scale
- Network effects
- Learning curves
- Price discrimination, bundling
- Game theory (prisoner's dilemma) !!!
- Market segmentation

Profitability Cases

Booz Allen & Hamilton - 1

Case scope provided by the interviewer

Branded consumer products manufacturer makes food for people and pets. Revenue and profits are stable only through acquisition. Otherwise, growth has slowed and revenues have been decreasing. One reason might be because of private label competition and ensuing pricing pressure. We are to concentrate on the dry pet food market – grain based (corn, wheat, soybean) – that is sold in grocery stores, mass merchandisers, and large pet stores (think Petsmart).

Additional information provided after relevant questions

- Concentrated on “Rolled-up” costs
    - COGS = 60%
    - Slotting Fee = 20%
    - Media Promotion = 10%
    - SGA = 10%
- COGS comprised of
    - Raw materials = 49%
    - Labor = 18%
    - Distribution = 21%
    - Packaging material = 12%
- Client ships from manufacturing centers to w arehouses around the country

Example - How to crack the case?

The interviewer was very stone faced throughout. It wasn't until the very end and I exhausted all possibilities of lowering costs did he show me a map of the U.S. with the client's manufacturing and distribution centers (DCs). The clients two manufacturing centers were located in Kansas and their DCs were located along the coast – the thought being it was more better to be closer to the raw materials. The interviewer told me there were regional manufacturing centers the client could rent or purchase, which would be lower cost than shipping the finished product.

Booz Allen & Hamilton - 2

Case scope provided by the interviewer

Client is debt-collection processing company and is concerned because it only collects approximately 30% of the bad debt that it's assigned to collect. What should the client do about this?

Conversational example - How to crack the case?

What are inputs into process and where are they sourced?
Client receives and tracks debt collection requests through a paper-based system. Some requests are received electronically but those are also converted to paper.

This seems highly inefficient; does the client have computer systems or databases with the capacity to store this same data?
Good question, yes they do, but their systems very by department, so they use paper – the one ubiquitous standard.

Regarding sourcing, does the client receive the debt collection requests from a few or many different sources?
They receive the data from many different sources – in fact, they have regional offices that source the data and then forward it on to a national processing center.

Does any conversion process need to occur when the data is initially sourced? In other words, do the regional processing centers transcribe the data into some standard form/formatting to transfer to the national office?
Yes, that's exactly right, they are responsible to putting the data into a standard format.

Do we have any sense about whether or not the individual regional offices format that data with varying degrees of quality or control?
Why are you asking that question?

I would like to know if the national processing center has to spend time further formatting any of the data they receive from the regional centers before actual processing can begin.
Okay, good question, but I don't have the answer to that. Why don't we think about the processing at the national center itself?

Okay, what can you tell me about the process flow for collecting the bad debts?
Can you be more specific? What would you like to know?

Well, I'd like to know what happens when the national processing center receives a bad debt notice? Do they attempt to make contact with the individual/group in question? How do they attempt to make that contact? Is there a case manager for each case?
Good questions. The national processing center reaches out to the debtors in a multi-stage process. First, they attempt to collect debt through mail, then, if there's no response after 90 days, they send another more urgent notice. If there's no response to that after 90 days, they attempt to call the individual and collect. Finally, if they have no success with phone calls, they have teams of people operating out of the regional centers who attempt to collect in-person.

Do we have a sense of the success/failure rates falling out of each of the stages in this process?
Yes, about 30% of people pay and respond to the first notice, about 40% of those remaining pay and respond to the second notice, about 20% of those remaining are reached via phone and, finally, about 10% of those remaining are reached in-person.

Interesting… intuitively, it seems surprising that the more personal contact results in a lower success rate. When they try to contact the debtors over the phone or in-person, what data are they using to make that contact?
They're using the personal information passed on to the debt-collection agency from the creditors through the regional centers.

So, given the window of time between when that information is passed along and when the first telephone contact is attempted, it could have been about 180 days, or six months! Given the nature of folks who don't respond to their first two debt collection notices, it strikes me that they may have changed their phone numbers and/or home addresses since that time.
Good point, that's possible. What else do you think could be happening here?

Well, going back to the data handling process, I would be concerned about the integrity of the data with it being tracked on paper and passed around from center to center. Does the data get passed on to different case teams as it moves through the various stages of collection?
Yes, it does, there are different case teams for each stage in the collection process. What else?

What about the creditors passing along the data to our client– how can we be sure that their information is very accurate. If they've had a difficult time tracking down debtors, it may be that person is not reachable at their old address. Does the debt collection agency have any kind of system for finding people based on their social security number or some other form of common identification?

Good question, no they don't - they have always relied on the data passed to them by creditors. Anything else?

That seems like a potentially limiting way of collecting debt, again, especially considering the probable nature of extreme debtors. What about the actual processing capability of the debt collection agency? Among all of the debt that they process, are they adequately staffed to handle the load? Is there debt that is not collected because it wasn't pursued?

Good thought – but we worry about that right now. Why don't you summarize your findings and let me know what you'd recommend as next steps for this agency?

Well, it's clear that this debt-collection company is running with an outdated system for managing data. Rather than using a centralized computer-based database for managing their data, they are using a decentralized paper-based system. In addition, they rely on their suppliers (the creditors) to provide them with the data and don't have the capability to ensure that data is accurate and up-to-date. This is reflected in the performance of the process, which only results in approximately 30% of the overall debt being collected. I would like to investigate the actual payment posting system a bit more before making recommendations. However, given the current situation, I would tentatively recommend a full process review, a centralized database for tracking and storing information and a system that enables the agency to get up-to-date address information for any individual or group who is passed on from creditors.

Nice summary, you might be interested to know that this was the discovery-phase of an engagement with a real client – the United States Internal Revenue Service.

Booz Allen & Hamilton - 3

Case scope provided by the interviewer

Your uncle has just inherited an oil tanker. He has asked you, as a Michigan MBA and wise financial advisor, to tell him how much this oil tanker is worth.

Interview Hot Tip: The solution of the case is not dependent on what your uncle plans to do with the tanker. Whether he plans to sell it or whether he wants to use it himself, the value of the tanker is equal to the NPV of the profits that can be extracted from its future use.

Conversational example - How to crack the case?

There are two methods to determine the value of the tanker:

1. Revenue generation capacity: This can be determined by assessing the demand & supply for the services of the tanker

2. Scrap value
That's good.

I would like to examine the revenue generation capacity of my oil tanker i.e. determine the cash it will earn.
Sure.

Firstly, I would like to understand the business. What kind of services does an oil tanker provide?
Very simple. It just transports oil from the rigs and dumps them at the shore.

Can you give me more details about the oil tanker? What kind of an oil tanker is this? What is its capacity?
Sure. Basically, there are three types of tankers: Large, medium and small. Ours is a small tanker. Large tankers can transport 1000 tons of oil, medium 750 and small 500 tons per trip. Assume each tanker makes one trip a year.

I would like to know the competitive landscape
There is a lot of competition in this business. There are 50 large tankers, 75 medium tankers and 50 small tankers.

What does the demand for oil tankers look like?
It's 100,000 tons per year.

This means that supply is greater than demand. Supply is 131,250 tons while demand is only 100,000 tons. To understand which of these tankers will service the limited demand, we need to understand the cost and pricing.

Ok. So, what do you want to know?

What is the cost incurred by an oil tanker in transporting oil?

That depends on the type of tanker. I will list it out for you:
Large: $50,000 per trip
Medium: $75,000 per trip
Small: $100,000 per trip
Assume there are no other costs.

That's interesting. Let me work out the economics of transporting oil for each type.

TYPE	Capacity	Cost of the shipping trip	MC of shipping 1 Unit of Oil
Small	1 Unit of Oil	$100,000	$100,000
Medium	2 Units of Oil	$150,000	$75,000
Large	4 Units of Oil	$200,000	$50,000

As oil shipping is a commodity, the price for oil shipping will be competed down to the marginal cost of production. The price for shipping 1 unit of oil will be brought down to $75,000. The Large oil tankers will still turn out a profit while half of the Medium tankers and the Small tankers will be driven out of the market.

Ok. What do you understand from this?

This brings us to some interesting conclusions. Since large tankers are most economical, the demand will first be fulfilled by large tankers, then by medium tankers and lastly by small tankers. Large tankers can fulfill demand to the extent of 50,000 tons (50*1000) and medium tankers can service the rest. Consequently, there is no demand for the services of small tankers.

Consequently, my small tanker is worth nothing except some scrap

Good.

Booz Allen & Hamilton - 4

Case scope provided by the interviewer

The partners at the Cleveland office bought some artwork 5 years ago. They now want to sell it. Your job is to figure out which channel they should sell in and how much they should sell it for.

Interview Hot Tip: Interviewer wanted to see how I think. Figure out the mix of artwork, list the most applicable channels per type of artwork in order of precedence, determine how to figure out demand, and set a price.

Additional information provided after relevant questions

- Artwork:
- 80% modern
  - 10% oil
  - 50 % mixed media
  - 5% sculpture
  - 35% prints

Example - How to crack the case?

- Purchase price sunk – remember this!
- Potential channels
  - Internet
  - Auction houses
  - Galleries
  - Cold call lists of museum frequenters
- Points to consider:
  - Customer elasticity
  - Popularity of artist
  - Availability of work
  - Location
- Potential buyers
  - Galleries
  - Professional offices
  - Auction houses
- Price will be determined by market considering everything listed above– no need to reach an exact figure.

Booz Allen & Hamilton - 5

Case scope provided by the interviewer

You are an advisor to Hannibal who is planning a voyage over the Alps to conquer Rome. How much food should he take when crossing the Alps?

Conversational example - How to crack the case?

I would like to determine how much food he and his group will need when crossing the Alps, and then I would like the sources of food that he will encounter in his travels.

To determine the amount of food I would like to look at the number of people and animals that he will be taking on his journey, I remember that he invaded with elephants. I would like to consider the different types of food that they will need. Then I would like to look at their consumption of food. Lastly I would like to consider the amount of time that Hannibal's band with traveling.
Hannibal will be taking with him 10,000 soldiers, 2,000 Calvary on horseback, and 40 elephants. You can assume people and animals eat the same food. People east 2lb per day, horses eat 10 lbs per day and elephants eat 50 lbs per day. They will take 4 getting to the Alps and 2 months crossing the Alps

They will take 6 months to cross the Alps. How much time to conquer Rome?
They expect about 6-9 months of battle

Do they need to worry about the trip back?
You can ignore the travel home

I am going to assume 30 days per month, and 6 getting across the Alps, 9 months of battle because of the magnitude of the problem if the army runs out of food. The army will require (10,000 people*2lb/person/day + 2000 people*2lb/person/day + 2000 horses*10lb/horse/day +40 elephants*50lbs/elephant/day)=44,200 lbs per day. They will be traveling (6months + 9 months)*30 days = 450 days. The total amount of food needed is 44,200 * 450 approx = 22 million lbs.

Now I would like to look at the sources of food. They will be traveling over the Alps, which I would expect food to be sparse. What about raiding villages before and after crossing the Alps?
Lets assume that we are friendly with the villagers that we pass before crossing the Alps and the villagers from the Roman empire will burn their fields and houses as we approach.

If we can't find food how much food can the people, horses, and elephants carry?
People can carry 50lbs, horses can carry 200lbs, and elephants are already carrying armaments so they can't carry any food.

Just from hearing the numbers it is not going to be possible for these people and horses to carry their own food. Is there another way to transport the food?
They do have the option of using a wagon to transport food. The wagon is pulled by two horses, driven by one driver and can carry 2000 lbs. In addition, the wagon cannot cross over the Alps

I would like to look at how much food a wagon can provide up to the crossing of the Alps. The horses and driver would eat 22lbs / day. 22 lbs * 6 months * 30 days =3360. The wagons aren't going to help the situation.

Is their any other option for carrying food?
No

They could take a large number of animals such as horses and use them are food. It could be set up in a large pyramid scheme.
Lets assume that can be done

In that case I would recommend that Hannibal seek out other sources of food, ways of carrying the food or attack Rome from a different route.

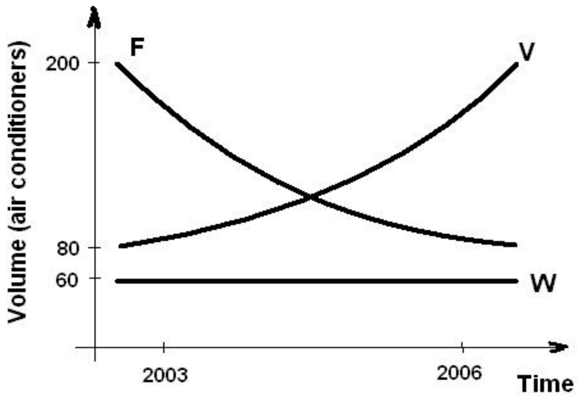
Booz Allen & Hamilton - 5

Case scope provided by the interviewer

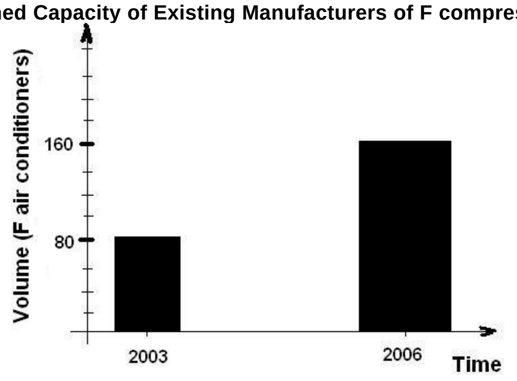
Your client is an industrial goods manufacturer who makes compressors for air conditioners. There are three types of compressors on the market: F, V and W, but the client only makes F compressors.

Until recently, the client had a dedicated customer in the form of its parent company, an OEM who makes air conditioners. However, a few months ago, the OEM divested your client. Now the client wants to explore whether it can sell to new customers.

Due to a major technological advancement, industry experts predict the following trends in the next 3 years in the compressor market:



Planned Capacity of Existing Manufacturers of F compressors



Analyze the situation and make recommendations to the client on how it should respond to these internal and external market changes.

Interview Hot Tip: This case is very broad. It could potentially be turned into a profitability or a market growth case. However, this particular interviewer had one concrete solution in mind. He only wanted to explore things from an economics point of view and after listening to my initial framework, immediately directed me to supply side issues. This shows well that some interviewers will have a very specific solution in mind; others will explore a wide variety of issues with you and just evaluate your structure and thinking. Ideally, you will come across both types in your 2-3 first round interviews. Sharpen your listening skills! Listening carefully and paying attention to the details is the only way you can tell where your interviewer wants you to go!

Additional information provided after relevant questions

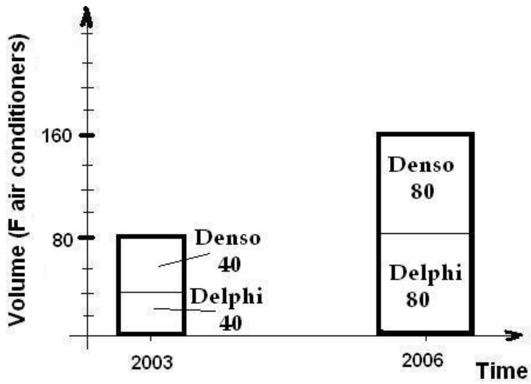
- Let's focus on supply-side issues only! Show me what will happen to prices in the current situation!

Key drivers of competition and comparison of compressor types			
Driver	**F**	**V**	**W**
Cost	Low	High	Low
Weight	Low	Medium	Low
Performance	Medium	High	Medium
Quality	Low	High	Low

- The new technology allows a company to make V type compressors at a Low cost by investing in a modern plant that makes 40 units of V compressors.

Competitor Analysis			
Company	**Location of markets**	**Types of compressors made**	**Comments**
The Client	North America	F	
Denso	Japan	F and V	Premium player
Sand	Europe	W	
Delphi	North Am. + Europe	F and V	

- The current and planned capacity of the existing manufacturers looks like this:



- Denso and Delphi, for whatever reason, are not able or do not want to add more capacity than what is indicated in the graph above.
- With the new technology, it is profitable to invest in a plant making 40 units per year, but it is not profitable if the plant operates well below capacity (e.g. 20 units a year). Raising capital is not an issue.
- The new technology is easily available to current market players.
- Sand is also considering entering the market for V compressors.
- Disregard logistics costs.
- Currently, V are more profitable than F compressors.

Example - How to crack the case?

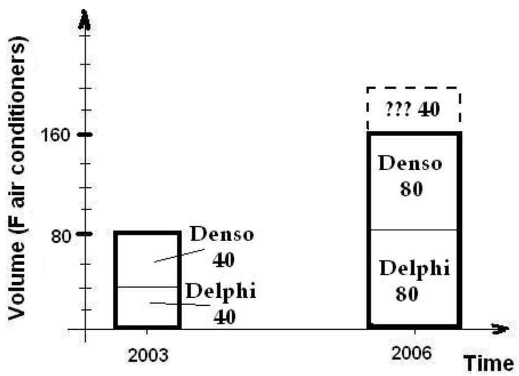
Suggested framework:

1. Analysis: What factors must the company consider in responding to the market changes?
   - Impact on profitability
     - Can sales growth be sustained?
     - How will our margins change?
     - Can the product mix be widened (diversify into V compressors)?
     - Investment needed to manufacture new products
       - New capacity/existing capacity
       - New technology
       - Economies of scale
       - Competitor retaliation (see below)
     - Changes in costs
   - Customers
     - Acquisition of customers new to the company
       - Costs of acquiring new customers (marketing, SGA)

        - Incremental revenues
    - Situation of existing dedicated customer (ex-parent company)
        - Potential for growth
        - Increasing competition?
    - Exploitation of new segments not yet exploited by competitors
- Competition
    - Drivers of competition
        - Price
        - Quality (reliability, etc.)
        - ?
    - Expected retaliation (see above)
    - New entrants (impact of technology on barriers to entry)
- Skills needed in changed market
    - Technology (potentially important due to new tech advancement)
    - Management skills
    - Marketing (probably not a point of focus in the past due to dedicated customer)
- Linkages between these issues
    - E.g. what skills do competitors have and what customers will they target

2. Recommendations

Based on the information above, we know that Delphi and Denso together will not meet the entire demand (estimated at 200 units) in 2006. There is an undersupply of 40 units as shown in the graph below:

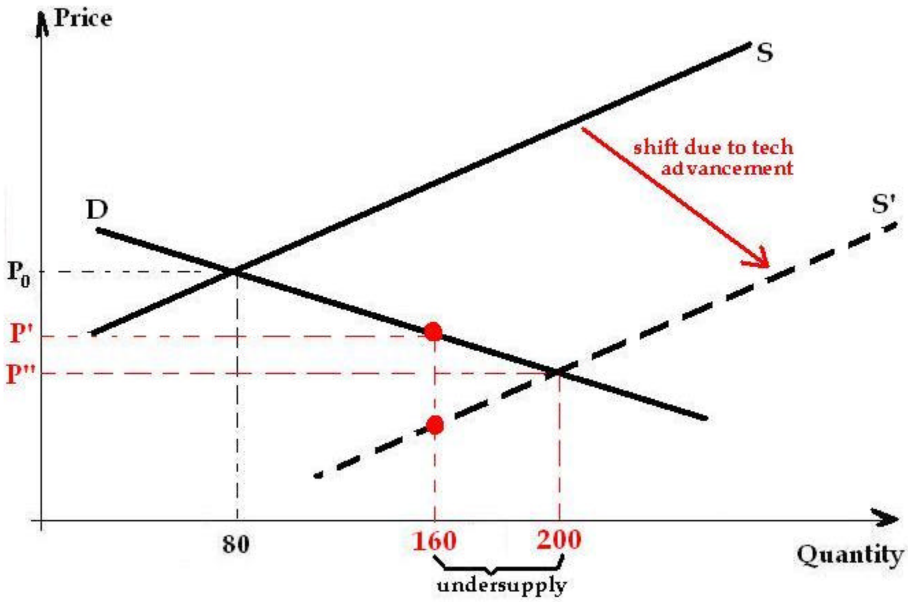


What will this do to prices of V compressors?

Currently, the market is in equilibrium with 80 units being produced and consumed at price P_0.

First of all, we know that the new technological advancement will lower costs for manufacturers of V compressors. Assuming that competition exists in this market, the supply curve will shift out as manufacturers are willing to supply a larger quantity at any given price. This shift explains the forecast that demand V compressors will rise from 80 to 200: customers want more of the higher quality compressors as their price goes down. (We assume that no major shift in customer tastes has occurred, leaving the D curve unchanged.)

However, we know that given this new supply curve, Delphi and Denso combined only plan on producing 160, while customers would demand 200. In this situation, undersupply will cause prices to settle at P', well above the equilibrium price of P''. This economic rent will only disappear if a new entrant comes into the market and produces the missing 40. Then a new equilibrium will be set at P_0 and 200.



Clearly, given the information we have about future trends in the air compressor market, the best strategy for the client is to enter the profitable and growing V market, especially since the F market, which the company serves, is projected to decline.

The only problem is that there is another firm, Sand, which will also see this opportunity. If both firms enter the market together, neither will be profitable, as the new plant is only profitable if it makes 40 units.

We can draw the following matrix based on game theory:

Game Theory Approach to Entering the F Market

	The client goes in	The client does not go in
Sand goes in	Both make losses	Sweet spot for client
Sand does not go in	Sweet spot for Sand	Nothing happens, Delphi and Denso get economic rent

The only feasible solution in such a situation is some form of collaboration. I recommend that Sand and the Client form a joint venture that will build a plant to make the missing 40 units, bringing to market into equilibrium in 2006 and competing away the economic rents that would have existed otherwise. Announcing such a joint venture should keep other new entrants out since they cannot make profits in this market.

Booz Allen & Hamilton - 6

Case scope provided by the interviewer

Our client, a financial services company sells credit cards. It has two major problems:

Dissatisfied customers are complaining about the level of service the company provides and that they have been treated unfairly.

The company has a hard time collecting money.

How would you analyze each of these problems?

Interview Hot Tip: This is an extremely broad case drawn from this interviewer's personal experience. Therefore, the interviewer will guide you in the direction he/she wants. The first step is always to set up a broad framework and then listen carefully to where the interviewer wants you to go.

Additional information provided after relevant questions

- The interviewer provided very little data, apart from a few initial remarks about how customers are approved and served. Then she just asked questions to guide me and gave 1-2 data points if necessary:

Conversational example - How to crack the case?

Part I:

Description of the customer management process (first she asked me how I think it is done and then she confirmed what I said and added a few points):

Before approving a customer for credit there is a risk assessment process, where the acceptability of the customer is determined. Then, if the customer is accepted, the client grants a certain type of card and services based on whether the customer is an individual or a company. Once the customer gets the credit card, he or she can use it at a number of partner locations (e.g. supermarkets, hotels, etc.). Client complaints are handled through a call center.

My conclusions based on this piece of information:

1. Risk

- Key issue: Are we having trouble collecting money because our clients have a high default rate, indicating that we are not evaluating risk properly?

2. Size and type of client (individuals vs. small and mid-size companies vs. large corporate)
   - Key issue: Do we have the resources and capabilities to serve the needs of our targeted segment?
3. Service level expected (wealth, type of purchases needed, etc.)
   - Key issue: What is the service our clients expect and why? How can we manage that?
4. Partner locations
   - Key issue: Is this the source of customer dissatisfaction? Are there enough partner locations and can the card be used quickly, easily and without problems?
5. Call center
   - Key issue: Is our call center effective at handling problems?

Interviewer: Ok, based on this, what are some reasons you can think of underlying the customer dissatisfaction?

Student:

1. Response time (how quickly are complaints handled once they reach the call center)
2. Mismanagement of expectations
   - Promise of ads/promotions does not match actual service
   - Service personnel
     - Lack of training
     - IT database needed
     - Miscommunication
3. Organizational problems (under motivated staff, not enough staff)

Interviewer: Let's suppose that you find that the core issue when it comes to customer complaints is that your staff in the call center is not able to respond quickly enough to complaints: How would you address this problem?

Student:

1. Optimize efficiency of call center (e.g. by pooling resources – one phone number instead of many)
2. Improve communication with other departments (cross-functional teams, etc.)
3. Improved training
4. Improve demand forecasts

5. Provide alternative methods of communication for customers (e.g. e-mail, self-help website)
6. Hire more staff in call center

Part II:

Interviewer: given that we don't have too much time, let's move on to the other issue: collections. You mentioned outsourcing as a key issue to consider improving collections. How would you evaluate whether to outsource?

Collections process (data given by interviewer after I asked the relevant question)

Customer does not pay on time ⇒ Reminder letter is sent ⇒ Person is given a telephone call ⇒ Collectors are sent (labor cost)

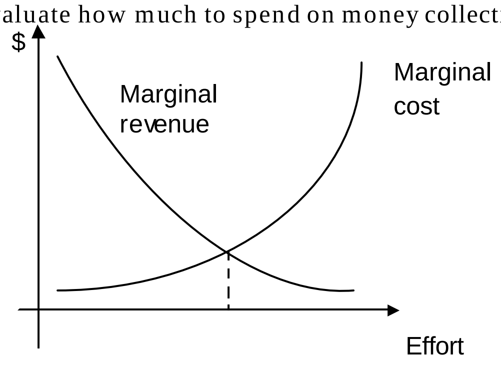
My analysis:
Profitability of collections = Revenue – Cost.

Collected money

Variable costs of collection:
-Labor
-Letters
-Telephone charges
-Etc.

Fixed costs of collections:
-Office space, etc.

How to evaluate how much to spend on money collections?



Resources dedicated to money collection should be increased up to the point where collecting an additional dollar would cost more than a dollar. Therefore the key metric to use for measuring the efficiency of collections is the cost to collect $1.

Interviewer: Ok, so let's evaluate how much the financial services company is now spending to collect $100:

Money collected: $2 trillion.

# of collectors employed: 100 000.

Salary and associated costs of 1 collector: $100 000/employee/year.

Student:

Revenue: 2×10^{12}

Costs: $1 \times 10^5 \times 10^5 = 10^{10}$

Cost to collect $200 is $1, so the cost to collect $100 is $0.5 or 50 cents.

Student: My analysis has addressed both problems our client is facing: customer dissatisfaction and money collections. I found that it is crucial to handle complaints quickly and effectively, so our slow response time is one of the main reasons why customers are dissatisfied. To decrease response time I recommend pooling resources at the call center, improving training and communication, hiring more staff, implementing systems to better forecast demand and/or providing alternative communication channels to customers. Based on my analysis of collections, we know that it costs 50 cents to collect $100 for the company and that the company would like to increase the amount collected. If other companies can collect more effectively for the same amount, I would outsource collections. However, outsourcing may not be the only way to improve collections. We need to benchmark this 50-cent cost to collect $100 with competitors' costs. Perhaps we are focusing too few resources on collection and that is why our collections are inefficient.

Booz Allen & Hamilton - 7

Case scope provided by the interviewer

A CPG company sells its products through multiple channels (specific product does not matter for this case). The majority of its products fall in the $.99 to $20 range. It advertises through media, print, radio, and TV. The CEO feels the effectiveness of the advertising and trade promotions are lessening. Why?

Interview Hot Tip: This case was much more of a conversation. I did very little writing and the partner did most of the talking. I thought this would be a very numbers oriented case, but it was more qualitative. Make an attempt at driving the conversation to where you think it should go.

Additional information provided after relevant questions

- Competitive landscape: store brands and private labels on the rise
- Customers are used to promotions and won't buy any other way
- Little differentiation among products
- Regional consolidation in mass merchandisers and grocery stores
- Advertising: consumers saturated with TV ads

Example - How to crack the case?

After a long discussion the partner told me the issue was how the regional sales managers were given incentives – they were given pricing power to make it more profitable to them and their channels while giving them leverage with the channels. i.e., $5.10<cleaner<$6.05

Booz Allen & Hamilton - 8

Case scope provided by the interviewer

The Vice President - R&D of a large CPG company, recently approached us. The VP had recently taken over and was very troubled by the fact that over the last 4 decades the number of new product launches as a percentage of total product launches had dropped considerably. Especially over the last few years, he had observed that most R&D resources were being focused on product modifications and upgrades rather than new product development. He strongly believes that there is a strong correlation between the company's drop in profitability over the years and the reduced focus on new product development. He has asked your help in validating his belief, identifying the root causes for the reduced focus on new product development and any suggestions for improvements.

Interview Hot Tip: First, I used the 3C framework to understand the market dynamics and the company's positioning. I used the framework to understand the changes in customer behavior and competitive dynamics over the years and to understand how the company had responded to the same. This helped me to validate the VP's belief about a correlation between new product development and profitability.

Next, I mapped the evolution of the company's R&D, the linkage between company strategy and R&D focus. I also identified the key factors that led to resource allocation decisions between new products and product upgrades.

Additional information provided after relevant questions

- The company had grown through acquisitions.
- R&D function was fragmented within the company. Product development activity was being undertaken at more than 7 R&D locations spread out in different parts of the world (wherever the CPG company had a large regional presence)
- Though there was a VP-R&D, the heads of various R&D setups reported functionally to him, but administratively to the regional heads of the CPG company
- There was limited knowledge sharing between the R&D setups
- The systems and processes for product development were different in each R&D setup
- Some regional R&D setups were heavily influenced by the corresponding regional heads of the CPG company, who in turn were driven by short term targets

- A major competitor had launched several new products in the past few years and due to this the CPG company had lost substantial market share
- Though the competitor had a de-centralized R&D setup similar to our company, over 8-10 years ago they had invested in a knowledge management system.

Example - How to crack the case?

- Lack a knowledge management system that can ensure effective sharing of knowledge across various R&D setups; there seemed to considerable duplication of effort and hence wastage of R&D resources
- Short term focus in some R&D units leading to increased effort in the direction of product modifications/ upgrades rather than new products
- Limited movement of scientists across locations; most remained in their local R&D units and hence limited cross-pollination of ideas
- Lack of metrics at a company level on allocation of resources for new product development as well as the number of new products
- Also, there were no specific incentives for the marketing and sales team to push new products over existing ones or to migrate existing customers from old products to new products

Booz Allen & Hamilton - 8

Case scope provided by the interviewer

The client is a company which specializes only on heavy equipment manufacturing. They serve two markets. The small equipment market consists of forklifts, landscapers, etc. The big equipment market consists of industrial machines worth over $2 million. The company is known for its premium quality for which it charges a premium price. It sells through a single channel: a network of exclusive dealerships that carry the full product line. It is considering implementing a CRM solution that would cost $60 million and would take 2 years to set up. Unfortunately, a pilot study, conducted recently by involving the dealers, was not well received and is judged a failure.

Is implementing a CRM solution a good idea? If so, why and what is the best way to do it?

Interview Hot Tip: This interview stressed the fit part more, so given that we only had 15 minutes left for this case, we did not go into too much detail with the interviewer. Given more time, I could have carried out a detailed cost-benefit analysis of the CRM solution, but we did not get to that. Lesson: Be prepared to sum up the case at any point, you should be able to provide a coherent synthesis of the main issues and a few recommendations to match.

Additional information provided after relevant questions

- The client has aggressive growth targets; it wants to grow sales by at least 20%.
- The client is a market share leader in both customer segments (big and small). It sees growth coming primarily from cross-selling opportunities and it saw a CRM solution as an ideal way of leveraging these cross-selling opportunities by sharing information across dealerships.
- The CRM solution collects information about customers in a large database using a uniform solution for all dealerships and all customers.
- The pilot study was used by the representative sample of dealers for a few months.
- The customers (the 2 segments):
- Small
    - 20% of the market, but growing.
    - Low margin
    - Served through local contractors where they go and chose from the wide array of products.

    - These customers mainly value price and selection although a few customers in the higher-end do value quality.
- Big
    - Mature market, 70-80% of the total market in share
    - High margin
    - Long-term contracts
    - Personal selling is the key to winning these big contracts, it is a typical "wine and dine" selling model.

Example - How to crack the case?

1. Understand why the client wanted to implement a CRM solution
2. Understand the CRM solution itself
    a. Main uses/benefits
    b. Reasons for failure of pilot study
        i. Problems encountered by dealers, etc.
3. Situation Analysis
    a. Customers
        i. Segments
        ii. Needs/Benefits
        iii. Interface method
    b. Competitors
        i. Drivers of competition (premium quality, service, etc.)
        ii. Benchmark CRM solutions, sales information management methods
4. Cost-Benefit Analysis of the CRM solution
    a. Costs
        i. Investment needed = 60 million + 2 years to implement
        ii. Additional resources needed (management time, new people, etc.)
    b. Benefits
        i. Forecasted incremental revenues for the client
        ii. Benefits for dealers
        iii. Customer retention
        iv. Better information management, leading to new customer acquisition

In evaluating this CRM solution I found that the client experimented with it in the hope that it would facilitate cross-selling between its customers and therefore help achieve its aggressive sales targets.

However, based on the information I have gathered so far, it seems that the failure of the pilot study was no coincidence. The larger and higher margin part of the market is all about **“wine and dine” personal selling**, where each new prospect deserves special attention and is a long-term investment. Dealers are likely to resent having to spend time with managing a costly database when clients are won through personal relationships not data mining. This is probably why the pilot failed.

Given this situation, the company should reevaluate its market position. Given that the high-margin business is stagnant and only the small segment is growing, pursuing **top-line sales** growth will lead to **lower profitability** in the longer term. The only way to increase market share in the **Big segment** would be to **invest in the personal relationships** with existing and potential buyers. Since we know that the **Small segment** mostly values **price and selection** over added services or quality, the client should focus on these attributes to leverage growth instead of investing in a CRM solution. Even if a CRM solution is pursued eventually, it **should not use a uniform solution** for the two segments given the different needs of the sales staff selling to the big and the small segments.

Booz Allen & Hamilton - 9

Case scope provided by the interviewer

Let's talk a particular trend in the US consumer industry, specifically concerning FMCG producers. I am referring to companies selling products between 99¢ and $20, like P&G, Pfizer, Coca-Cola and Unilever. The trend I have in mind is that A&P (advertising & promotional) spend is becoming less effective for these producers. Under advertising I mean communication through the radio, TV and outdoor channels, while promotion is about communication through the retail channel (merchandising, price discounts, coupons, weekly ads, etc.).

There are 3 questions I want you to answer:

1. How would you define the effectiveness of A&P spending?

2. Why is A&P spending becoming less effective in the US consumer industry?

3. What would be your advice to acompany in this industry?

Interview Hot Tip: Since this is a 2nd round case, it is much more strategic in nature, so the interviewer was testing the range of issues I was aware of, but also wanted to know my depth of knowledge on individual issues. Getting the definition of A&P spend effectiveness right seemed to be key. He always tested me to make sure I was confident in my responses, when he was happy with a framework I used, he jumped immediately into one of the sub-issues. I was pretty happy with the interview because I felt I had covered a lot of issues, but as a feedback they told me I could have gone into more detail on some of the individual issues, so it is important to find a balance between depth and breadth.

Conversational example - How to crack the case?

What is A&P effectiveness?

I provided the following definition:

- $ increase in profit / $ spent on A&P
- but suggested that there could be other definitions, such as:
- recall: % of target group who recalls the ad or promotion after a given time period
- reach: % of target group reached by the ad or promotion
- long-term effect on brand equity.

The interviewer wanted to know why I didn t use *$ increase in sales / $ spent on A&P* instead of *$ increase in profit / $ spent on A&P*. Because we always have to consider both the increase in sales and costs, an ad or promotion is not effective if it generates sales growth, but fails in increasing profits.

Why is A&P spending becoming less effective in the US consumer industry?

My suggested framework of issues:

Share of Voice (SOV**)** is becoming more expensive to buy as individual companies increase their spending and competition increases

Reasons for this:

- Threat from private labels on the low end as the power of retailers increases
- Threat from "new luxury" items on the mid-high/high end (see BCG article on "Trading Up") as the middle class is able to afford higher-priced everyday items (e.g. wines)
- Markets reach maturity so competition becomes fiercer
- Globalization: new entrants from foreign markets into the US
- Change in media channels
- The effectiveness of TV is decreasing as consumers substitute other channels such as the Internet
- Consumers collect information from more channels (increasing "noise")
- Reasons internal to the individual **companies**
- Demographic trends (e.g. growth of sub-groups in the population who are reached less effectively through traditional channels (e.g. Hispanics))

(After going through my framework, we talked in depth about three issues: growth of private label products, growth of "new luxury" items and the effects of globalization on the US consumer industry. He did not give me any data, he just asked me to show how each of these issues impacted A&P effectiveness.)

What would be your advice to a company in this industry?

My suggested framework:

- Use an analytic model to identify effective A&P activities as outlined in the presentation given by Booz Allen Hamilton in 2002 Winter at the business school (see MCC website for complete presentation). This method can be used to identify best practices inside the company and scientifically identify what made a given ad or promotion successful, so this can be repeated in the future. An important component of this model is empowerment of the sales force to make or contribute to decisions regarding A&P activity in their respective territories.

- Address organizational issues that may be hindering successful A&P activities
- Ensure better and faster communication between the sales force and management
- Optimize sales force structure
- Find new ways to communicate to selected target groups
- Benchmark against the competition, seek out best practices outside the company

(At this point, time was running out, but the interviewer had one important issue in mind that I hadn't mentioned.)

Let's say you implement all the ideas you outlined above and there is still no increase in A&P effectiveness. What is the key issue you must consider?

The answer he was looking for was that the sales force compensation system had to be aligned with the goal of increasing sales force effectiveness. Based on the definition I gave, rewarding the salespeople based on sales targets will not necessarily increase A&P effectiveness, as the sales force will push even for expensive local promotions that are not worth their cost but increase sales.

Bain & Company- 1

Case scope provided by the interviewer

Our client is a big food retailer like Applebee's. They have 1000 restaurants, and 75% of them are franchises. The average revenue per restaurant is $2M and the average operating profit is 25%. Our client believes that the franchises are performing better than the company owned restaurants. Our client aims to find some ways to increase the overall profitability by $40M in one year. How can our client achieve this profit increase?

Conversational example - How to crack the case?

There are two main ways to increase the profits.

- Enhance revenues
- Cut costs
- We should look at these items and compare the averages of company owned restaurants' with the franchises'. Then, we need to adapt the best practices to our client's restaurants. Let's start with the costs first: I would like to look at the main cost items like direct material, direct and indirect labor and other types of overhead costs. What are the main cost buckets?

(Note: Bain interviews are different. When you ask the right question, the interviewer presents you with some slides that incorporate the information you are looking for.)

(Here, the interviewer showed a slide with all cost items and their monetary values. It is important to start investigating the cost cutting opportunities starting from the biggest cost items.)

- I found out a food cost reduction opportunity that cost the company $15M to implement. (They had problems with inventory storage and could solve it through a new IT system). The breakeven would be around 3 years. And the NPV of cost savings was $100M.
- Then I looked at the labor costs and found that there was a $25M cost saving opportunity.

The interviewer asks in between why there might be a discrepancy between the performances, and you need to come up with ideas like: productivity of employees, employee/customer ratio in the restaurants/ incentive system etc.

- Then I looked at the revenues by comparing the average revenue/restaurant and found out that there was a 12.5M revenue increase opportunity.
- Our client should definitely go ahead and implement the cost savings and revenue enhancement opportunities. In one year our client can only achieve 25M+12.5M=37.5M of profit increase, but if they implement the food cost reduction system, in around 3 years their profits will increase even further. So, they should definitely go ahead and employ the new IT system, as well.

Bain & Company- 2

Case scope provided by the interviewer

Your client is a retail bank, which has an area of Private Banking, thatacquired an investment banking that also has a Private Banking area. What should the client do with the 2 areas?

Interview Hot Tip: Merge if there are synergies / economies of scale, restructure / sell / close one if there are diseconomies of scale or other problems.

Additional information provided after relevant questions

- No opportunity to reduce costs in purchasing depot (small volumes).
- No best practices from one bank could be applicable to the other bank.
- Percentage charged from client as 5% and from target bank was 3%.
- Clients from the investment banking (large pension funds) would not agree to pay 6% commission. Previous clients would want to pay only 3% commission if the operations were consolidated.
- Revenue from client = $320M. Acquisition target's revenue = $220M.
- Investment bank charged only 3% commission because once they brought the pension funds to this kind of investment, they could refer some of them to other products.
- Avg Revenue per client = $250k
- Selling Price of a client to another bank = $80k
- Margin of the business = 8%

Conversational example - How to crack the case?

- I went straight to the value chain to identify opportunities to reduce costs. There were no opportunities. So I chased opportunities to increase revenue but percentages charged were different (impossible to increase from those who paid less – large pension funds). The problem is, the client would be blocked from raising percentage charged, and also has to reduce from previous clients (no price discrimination among clients would be possible).
- No cross-selling opportunities, same products were sold on the 2 areas of the 2 banks. Therefore, merge them was not an option. There were no opportunities for restructure as well.
- Sell the investment bank's division (client base) to another bank could be an option. After calculating the sales value I realized it was the best alternative.

Bain & Company- 3

Case scope provided by the interviewer

Our client is Cement Corporation, their biggest competitor id United Corporation. Our client's revenue is twice as much as United's although it's profit is half of United's profit.

Conversational example - How to crack the case?

After I learned that for the sake of the case the product mix is the same for the two companies, I've set up my framework: Profit is a function of revenue and cost, since our revenues are twice as much, it looks like there is a very big cost problem here. Let's look at the cost items and compare them with United's.

(The interviewer did not let me go away with costs and pushed me a lot about the revenues and wanted me to come up with various reasons why there might also be a revenue problem. Actually, he was just trying to give me a hard time and see how I was reacting to his behavior and whether I could come up with differentideas.)

Which cost items do you want to look at?

The variable costs such as Direct Material, Direct Labor, Processing costs and Fixed Costs such as Depreciation and SG&A .

Which ones do you think are the most important ones?

Since this is a manufacturing operation I would expect the direct material to be a very important part of it. How many plants do we have (-6), how many does United have: (-3). So, I think SG&A, and especially distribution might be very important.

(Note: He pushed me until I came with really reasonable answers about why these costs might be different for the two companies. And he did not give me any data until I found the exact reason for why the distribution was an important factor.)

I figured out that since cement is a heavy material the easiest and cheapest way of transporting it was using ships. Then, I asked if our customers and United's customers where in the same locations. I hypothesized that our customers might have needed ground transportation where the competitor's could be close to river or sea. Which was exactly the case, and the interviewer showed me a slide showing the plants and customers of our client and United.

- Our client needs to segment the customers according to location and charge them premium price if they need ground transportation because they are in-state.
- Our client needs to look at the profitability of each plant and close some of them foregoing some of the clients if necessary. That will decrease the depreciation costs, as well.
- After improving current distribution, our client needs to look at the other cost buckets and try to find some cost cutting opportunities.

Bain & Company- 4

Case scope provided by the interviewer

Our client is in the printer and cartridge business. They operate in 5 different printer segments according to quality and price of the printer. They are the market leader. Our client believes that company has become lazy and there is some value within the business that needs to be extracted.

Additional information provided after relevant questions

- The profitability and the market shares in each of the segments, the competition, the evolution of the competition, evolution of the size of the segments

Conversational example - How to crack the case?

(Note: In Bain cases it is very difficult to come up with a framework and carry it until the end of the case. Usually, they give you some hints throughout the case and they expect you to listen to them and adapt your structure to it.)

- We find out that a competitor is selling the printers with a loss in the low quality segment and basically driving our client out of business. They are making the whole money from cartridges. Cartridges are also cheaper than our client's cartridges, since the competitor has licensed the cartridge business to a Japanese company that can produce them for a very low cost.
- When we look at our cost structure, we see that whatever we do we cannot compete with the competitor in cost.
- The whole printer business is migrating towards more quality and more speed, and these areas are where the profit margins are much higher as well. So, our client should exit the Segment 1, slowly leave Segment 2 as well, and invest all its resources to coming up with even better quality products. If they go to a price war, our client will definitely lose the war and a lot of money.

Bain & Company- 5

Case scope provided by the interviewer

Your client is a large railway that is not yielding the expected returns. How can you turn-around the situation?

Interview Hot Tip: Search for opportunities to increase revenue and reduce cost. Search for potential acquisition.

Additional information provided after relevant questions

No info was given, I had to structure what information I would get and what would be some hypothesis of opportunities to turn-around the railway

Conversational example - How to crack the case?

- I focused the case on opportunities to increase revenue. I would first identify all the types of cargo transported by our client and the respective revenue. It would be the baseline for improvement. Then I would map what are all the large producing items on the region and identify whether those items (steel / woods / fruits / etc) are shipped or not by our client.
- The 1st opportunity would be to increase market share with items that are already transported by reaching clients. With clients that never bought our service, the sales force should reach them and present the opportunity to ship via the railway (economic model comparing costs of transportation and services of our railway against competition). For lost clients I would understand why they left and address the issues to re-gain clients.
- The 2nd opportunity would be items that are not shipped yet. I would model the cost for the end-client of transporting in our railway against using the other alternatives (trucks or other railways).
- For all the opportunities an NPV analysis would tell whether the opportunity is worthy to be pursued.
- In the cost side I would breakdown all the value chain and search for opportunities of improvement in the actual processes (gain processes efficiency), in the actual work force and in the actual assets (outsourcing). A benchmark with costs of other railways could also be done to identify where the client has more room to improve.

Miscellaneous

Case scope provided by the interviewer

ABC Power Company is an international developer that builds and acquires power plants. The industry had been growing well. However a recent slow down has caused an overbuild situation and the company is strapped for cash. How can we get out of the liquidity crunch?

Additional information provided after relevant questions

- Energy sources are commodities and the industry is growing at 2-3% per year.
- Types of plants- 80% coal, 10% gas fired, 10% hydro plants
- Operation- 300 days, 20hours/day
- FC- IT, Plants, and utilization
- VC-raw materials, labor

Example - How to crack the case?

- How big a liquidity crunch? Do they need money tomorrow?
- Still profitable but declining?
- What is the forecast for long-term growth?
- How can we cut costs today and raise cash?

Possible ways to solve cash flow:

- Reduce price of raw materials
- Factor receivables
- Sell off unproductive assets
- Increase utilization through outsourcing
- Labor reduction?

After framing the industry and company and breaking down where the FC and VC cost are in the industry, interviewer provides information regarding the coal part of the business:

The company is looking at possible purchasing options for its coal power plants and has been given a couple of options. They can either purchase mixed coal (option 1) or can purchase it type (option 2). In both cases they will utilize the following volumes (as a percentage of all of their needs).

- Purchases 3 different types of coal:
  - 50% Standard- 100K Tons/ hr

- 30% Better - 80K Tons/ hr
- 20% Premium- 40K Tons/ hr

- Option 1:
  - $10 for the first 400K
  - $8 for every additional ton
- Option 2:
  - $400/hour Standard
  - $1000/hour Medium
  - $1500/hour Premium
- Cost for option 1: 300 Days x 20 hours= 6000
- Standard requires: 3000 x 100= 300,000
- Better 6000 x .3=1800 x 80 = 144,000
- Premium 6000 x .2 = 1200 x 40 = 48,000
- Total coal required= 492,000
- Cost

400,000 x $10= $4,000,000
92,000 x $8 = $736,000
Total= $4,736,000

- Cost for Option 2:
- Standard: 3000 x $400= $1,200,000
- Better: 1800 x $1000= $1,800,000
- Premium 1200 x $1500= $1,800,000
- Total Cost: $1,200,000 + $1,800,000+ $1,800,000= $4,800,000

Take option 1

Following case question provided by the interviewer

The company is also looking at selling off some of its assets. It has three possibilities, which is the best asset to sell?

Example - How to crack the case?

- Three assets:

Asset 1- Future cash flows from the plant are $300 M

Asset 2 – Future cash flows are $200 M

Asset 3 – Future cash flows are $200 M

- If you sell asset 1 the company will get 60% of the FCF value and will pay 20% tax (5 more years of operation)

- If you sell asset 2 the company will get 75% of the FCF value and will pay 5% tax (3 years of operation)
- If you sell asset 3 the company will get 90% of the FCF value and will pay 10% tax (1 more year of operation)

Asset 1 $300M x 60%= $180 M - $36M= $144 M

Asset 2 $200M x 75%= $150M – $7.5M= $142.5M

Asset 3 $200M x 90%= $180M- $18M= $162M

- Sell Asset 3 if you cannot utilize it for other purposes to get better value. Another possibility is to learn about the efficiency of the plant, can it export some of its energy or excess capacity at higher prices?

Miscellaneous - 2

Case scope provided by the interviewer

You are on an airplane coming back from an engagement. About 50 minutes before you land you start a conversation with the person next to you who happens to be the CEO of a Pharmaceutical company. He learns you are a consultant and asks your advice regarding his oncology division.

The Cancer Division is a $2B a year business with sales increasing at 15% per year. It has three products A, B, and C. Product B is coming off patent soon. Does the company have any concerns? Should it make any changes?

Additional information provided after relevant questions

- Products:
- A -20% of business
- B-30% of business
- C-50% of business
- A is growing at 30% per year
- C is decreasing at 10% per year

Example - How to crack the case?

What is B's sales next year, growth rate, and % of total division?

Solution for part A:

- Present sales= $2B
- Product A= 20% or $400M
- Product B= 30% or $600M
- Product C= 50% or $1B

Next year sales:

- Total sales increases by 15% so $2Bx1.15= $2.3B
- Product A is growing at 30% so $400M x 1.3= $520M
- Product C is decreasing at 10% so $1 B x .9= $900M
- Total sales for B= $2.3 B- $520M-$900M= $880M

Product B

- Last year= $600M this year =$880M
- Increase =$880/600M= 46% increases
- % of Total= $880/ $2,300= 38%

Knowing that the B product will be coming off of patent should be a big concern. The company needs to react, some possibilities:

- Do they have a new product to replace this product in R&D?
- Can they create a partnership with a Biotech to develop a new drug?
- Can they in-license a product from another biotech or pharmaceutical company?
- Can they diversify into other areas?

Part B:

The company does have the potential to in-license 2 drugs:

- One is a late stage product- product D
- One is in clinical trials- product E

Additional information provided after relevant questions

- If they in-license product D there is a 60% potential they will get $2B next year forever
- If they in-license product E there is a 30% potential they will get $4.5 B starting in 2 years forever
- Discount rate is 15% but only provide if requested.

Example - How to crack the case?

Product D

- Expected value is $2B x 60%= $1.2 B
- In perpetuity= $1.2 B/.15= $8B
- Need to discount this back 1year= $8B/1.15=$6.95 B

Product E

- Expected value is $4.5 x 30%= $1.35B
- In perpetuity in 2 years= $1.35 B/.15=$9B
- Need to discount this back 2 years= $9B/(1.15)^2=$6.8 B

So, go with product D!!!

Cargill -1

Case scope provided by the interviewer

You are a part of the SBD group and receive a call from the R&D. They have been working on a developing crotonoids (anti-oxidants) using biotechnology. Currently crotonoids are used in three different markets: 1) Animal Feed, to add color 2) As a preservative to extend shelf life, and 3) As fortification in natural food market. Currently there is a market for crotonoids and it is serviced by companies, which provide chemical/synthetically-developed crotonoids as compared to the natural way of producing it as proposed by our R&D. It is estimated that it will cost about $5-$7 million over 5 years to develop a ready to market crotonoids.

We would like to know if it is worth pursuing this investment in R&D? and in which one of the three markets (you can only pick one)?

Interview Hot Tips: This is a broad strategy case. Analyze the size of all three markets first by asking about growth patterns, competitive landscape, different products and customers. Ask for any new potential entrants and as well as shifts in competitor strategy. Then proceed to inquire about why Cargill is interested in this market, both for the short-term and long term.

Conversational example - How to crack the case?

(After discussing the framework with the interviewer…) Let me start by understanding each of the three markets first. Let's start by understanding the Animal Feed market. Can you tell me little bit more about the market in terms of its: size, segments, growth, and uses of crotonoids?
Sure. There are two primary segments in the market: 1) Beef producers and 2) Fish Farms. Beef producers use anti-oxidants (crotonoids) primarily for food coloring and it's about a $1B market with no growth. Fish Farms is an embryonic market for crotonoids and this market also uses them for food coloring. In the Fish Farm market as well the growth is flat.

Good. To complete my understanding of this market, I would like to know who the competitors are as well as what are the value drivers of the customers are for the Beef Producers and Fish Farms.
Competitors A & B control about 95% of the market. They currently provide a chemical (artificial) substitute in this market. Both are very well entrenched and have been supplying to both the segment for the past 30 years.

As to the second part of your question, the content of the anti-oxidant (crotonoids) is so small that it is not even listed as a part of the ingredients. Given this information how do you think customers will value a "natural" crotonoid?

Based on the information you have given me, I would assume that the end users of beef and fish farms do not value a "natural" crotonoid. Is that a correct assumption?
Yes.

So is it also correct to assume that Beef Producers and Fish Farms will not be able to differentiate their products by buying a "natural" crotonoid from us?
Correct.

Ok, so this information tells me that we cannot sell our crotonoid at a premium versus the competitors' products. Now can you tell whether we can produce our crotonoid at a cheaper than Competitors A or B?
Actually our manufacturing costs are higher than competitors A & B.

All right. Based on the information available so far, the Animal Feed market does not seem to be an attractive target to enter. However, let's look other markets before we can say that with finality.
All right. Which market would you like to know more about next?

I would like to know more about the preservatives market. Can you tell me little bit more about the market in terms of its: size, segments, growth, and uses of crotonoids as well as who our competitors are in this market?
Preservatives market is about $100 M market with competitors A& B dominating this market as well. Similar to the additive market, the usage of crotonoids is small enough that it does not even warrant a mention in the list ingredients on the label. For the sake of this discussion you can assume that the market is not segmented and growth is flat.

I see. It seems that the preservatives market displays very similar characteristics as the additive market. Based on the information we have even the preservatives market does not seem attractive. Let's look at the natural food market.
Very well, what would you like to know about the natural food market?

Can you tell me little bit more about the market in terms of its: size, segments, growth, and uses of crotonoids as well as who our competitors are in this market?
Natural food market is also about $100 M with competitors A & B dominating this market as well. Before I answer other questions, why don't you tell me, based on your information, about the growth and segments for this market.

Based on the current trends about the benefits of natural foods and the growing interest in consuming natural goods, I would say that there is significant growth in this market as well as a particular segment of consumers who prefer these types of foods.
Very good. Your analysis is correct. Natural food market is constantly striving to achieve a 100% "natural" content in all the products it sells.

Now I would like to understand what the financials of entering this market. Can you tell what: a) price of crotonoids, b) our cost-basis, and c) competitors cost-basis
For the sake of this discussion you can assume that it is not profitable for us to enter a market as small as natural foods.

All right. Let's try to understand the competitive landscape. I would like to know if: a) Competitors A or B are working on developing a bio-tech version of crotonoids, b) Any other companies planning to enter crotonoids market, and c) Size and history of competitors A & B.
Neither competitor is working towards developing a bio -based version of crotoniods nor are any other companies planning to enter this market (as far as we know). Competitors A & B have been around for past 30 years. Competitor A has about 75% of the market, and by far is the biggest player. Both have established significant relationships clients in all the three markets.
You have almost all the data you need, why don't go ahead and summarize the case.

In exploring the feasibility of entering a market for bio-based crotonoids the data points towards entering the natural food market. However, small size of this market is a deterrent in entering this market.

Based on the data provided, it is my recommendation that we enter this market because:

- There is growing want on part of consumers to move towards natural foods.
- We will develop economies of scale and scope to reduce our cost basis
- It will build our expertise in developing bio-engineered products which can be applied to developing other products.
- It is also recommended to enter the market in the form of a JV with Competitor A. Thus we can leverage their relationships and they can leverage our technical expertise.

Very good. Thank you.

McKinsey & Company - 1

Case scope provided by the interviewer

Our client is Metrix Corporation. They are an electronic device company. They are laptop-desktop manufacturers and are selling to retail electronic stores. Our client considers to enter the PDA market.

What are the key areas to investigate?

Conversational example - How to crack the case?

Interview Hot Tip: The most important thing is to see whether the NPV of this market entry will be >0. I would look at the Market (size/growth/competitive landscape/market shares/barriers to entry); Customers (type/segments/retailers/distribution); Company (core capabilities, distribution channel, capacity, financials)

Let's focus on competitive landscape. What do you want to know about that?

- Who are the main players?
- What are their market shares?
- What other businesses are these players in?
- What is the cost structure?
- What would be the competitive reaction?
- What are the differentiation possibilities?
- Mode of entry and possible acquisition targets

In 96, there were 3 main players with 40M out of a total market of 50M.

In 99, the main players had 150M out of a total market of 250M.

- What is the % change in market shares of these players; and the total market?

Market share: 40M/50M=80%, 150M/250M=60%, 80% to 60% > 20%/80%=25% change

Total Market: (250M-50M)/50M=400% increase in 3 years = 133% increase/year

- What does this information tell you?

That the market is growing very rapidly, but the number of small players is increasing even quicker. Most probably these new entrants are driving the prices down.

In 96, the price of one PDA was $500.

In 99, the price of one PDA had dropped 75%.

So, the price has dropped to $125. So, the PDA market was 50M/500=100K units in 96, and in 99 this number had increased to 250M/125=2M units. So quantity in the market is increasing and the prices are decreasing.

- How many units should our client sell to breakeven in 3 years?
- What is the Initial investment necessary?

$150M

- What is the VC/unit?

$75

If we take the p=175, and discount rate to be 0,

(175x-75x)*3 = 150M

x=500K

(Here, the interviewer showed me the major players and their sales projections, and the major ones were planning to sell around 500K, also the major players had the synergy of being laptop-desktop producers.)

- Our client should not enter this market. Within the giving pricing structure, it is already very difficult to breakeven. Looking at the history, it is obvious that the prices will increase even more, decreasing the profit margins.
- The only reason our client might enter this market is, if there are any synergies that will decrease their variable costs down, or that will result in increased sales in their other products.
- How would you convince a client that does not want to listen to you?

I would follow the client through my analytical process, show him/her all the supporting data/ talk about the synergies to investigate and finally, show what competitive actions might come into place.

McKinsey & Company - 2

Case scope provided by the interviewer

Our client is a hair color company. Their products are hair color 45%, hair treatment 40%, other 15%. Their market share has been falling down. What could be the reason?

Conversational example - How to crack the case?

- After I identified the issues (customer segments, promotion, distribution channels, competition, etc.) the interviewer gave me some data that showed that in the men segment we have 20M out of a total of 225M, and in the teen segment we have 30M out of 90M, in the women segment we have 400M out of 825M.
- The men and teen segments are the fastest growing segments and we are very poor in these two segments.
- Here, the interviewer wanted me to calculate "How much market share should we have in the women segment, in order to have a total market share of 50% 2 years from now with the growth rates of 20%, 10%, and 5% in the men, teen, and women segments respectively?"
- This looks easy, but the calculation is pretty long and complex and you need to simplify your calculation in order to finish it, and also impress your interviewer.
- Then, the interviewer showed me some data about the awarenessof our client and the biggest competitor. The awareness and quality ratings of our competitor were better both from the user and non-user perspectives.
- Our client needs to revise its brand image and invest in advertising. Most probably our client did not a dvertise for a long time, and only the old consumers know its quality. So, our client should advertise, and try to attract new customers. Furthermore, our client should invest in the men and teen segments that are growing so rapidly.

McKinsey & Company - 3

Case scope provided by the interviewer

Our client is Travel time, a travel membership and discount company. Their revenues come from annual fee and commissions. The number of their members has been dropping resulting in a 60% revenue decrease.

What could be the reason for the membership decrease?

Conversational example - How to crack the case?

- Competitors (There might be a new entrant into the market, the existing competitors might be giving better discounts or might have lower annual fees),
- Direct sales (Customers might no be using travel companies any longer, and might be getting the discounts from the travel companies directly.
- Company (Company might not be reaching well to the customers, the customer service might be poor)

The annual fee our client charges is $60, and the market average is $50. Every year, out of 1M new customers, 60% are expected to leave. If our client drops its fee to $50, 1/3 will stay. How much money would our client make in 2 years(if we take this moment as 0)?

- If it charges $60: 1st year: 1M*60=60M, 2 nd year: 1M*0.4=400K 400K*$60=24M, in total 84M
- If it charges $50: 1st year: 1M*50=50M, 2 nd year: 400 K + 600K/3=200K will stay, so 600K*$50 = 30M , in total 80M
- So, our client should not drop its price to $50, but instead try to increase the effectiveness of its communication to its current and potential members. It should properly communicate its members the companies that are in our client's portfolio, the benefits its members can get and so on.

McKinsey & Company - 4

Case scope provided by the interviewer

Imagine you are in the UK in 1997. A retail clothing company sells mid-range young professional clothing to men and women through retail stores and catalogues. From 1987 to 1995 catalogue sales grew rapidly – typically, catalogues were a subset of the total number of items sold. Since 1995, catalogue sales have been flat or declining. The CEO would like to know why and what can be remedies can be installed.

Interview Hot Tip: As with most McKinsey interviews, the interviewer will give you very little time to sketch out a framework. I did have enough time to draw out a revenue tree and a cost tree. He drove the conversation from here.

Additional information provided after relevant questions

- 2 million catalogues mailed/yr
- Fixed costs = $6M/yr
- Variable costs = $10
- Price = $35 (average order)
- Average order is six items

Example - How to crack the case?

- The interviewer saw that I broke the costs up and he asked me for specific fixed costs and variable costs– remember to list the most relevant ones to the case first!
  - FC
    - Selling – printing costs
    - General – n/a
    - Admin – n/a
  - VC
    - Wharehousing costs
    - Buying of clothes from manufacturer
- Break Even volume = $6,000,000/($35-$10) = 240,000
- Hit Rate = # orders/catalogues shipped = 40,000/2,000,000 = 2%
  - 240,000 catalogues/6 items = 40,000
- Improve profitability
  - Increase price
  - Decrease catalogues
  - Decrease clothing costs

- Remedies
  - Scrap catalogue business with advent of internet
  - Specifically target catalogue mailings to improve hit rate, thus improving profitability

McKinsey & Company - 5

Case scope provided by the interviewer

Big Power Company owns and operates 100 plants in 25 countries. The company grew by building and/or acquiring plants in recently de-regulated markets. In the past few years they have seen their liquidity ratios fall to unacceptable levels. The CEO has asked our firm to look at three things: 1) improve their cash position 2) improve their cash reserves ($1B in 12 months) and 3) reduce the amount of debt Big Power is carrying.

Interview Hot Tip: The interviewer didn t give me time to lay out a framework or structure despite my repeated attempts to do so. Rather, he had a series of questions to ask me. He wanted me to brainstorm three or four solutions to each question. He would pick the suggestion he wanted me to explore further. Remember to think about your answers – list from most important to least important.

Example - How to crack the case?

- Question 1 -- What are some ways Big Power can generate cash?
  - Sell s maller and/or inefficient plants – Interview made note of this
  - Induce commercial customers to pay earlier– helps accounts receivables
  - Sell accounts payable to a third party
  - Re-negotiate loans
  - ** Cannot affect prices since energy is a commodity product **
- Question 2 – How can Big Power improve costs?
  - Reduce material costs, especially coal – Interviewer made note of this
  - Employ latest technology
  - Reduce labor costs
- Question 3 – Calculate the average cost of coal burned in one year
  - Operate 300 days/year
  - 20 hours/day
  - $10/ton up to 400,000 tons
  - $8/ton thereafter
- Ask about coal mix
  - 50% standard = 100 tons/year = 10 hours/day
  - 30% superior = 80 tons/year = 6 hours/day
  - 20% premium = 40 tons/year = 4 hours/day
- Question 4 – Should Big Power alter their current coal purchasing contract
  - $400/hr for standard

- $1,000/hr for superior
- $1,500 for premium

- Question 5 – Options to raise $1B in 12 months
  - Selling small/inefficient plants
  - Sell excess capacity
  - Raw material expense
  - Labor
- Answer to Question 3
  - Standard = 1,000 tons
  - Superior = 480 tons
  - Premium = 160 tons
  - Total = 1,640 tons for 300 operating days per year = 492,000 tons/yr
  - 400,000 tons * $10/ton (first 400K tons) = $4M
  - $8/ton * 92,000 tons = $736,000
  - Total cost of coal = $4,736,000
- Answer to Question 4
  - Standard = $4,000 ($400/hr * 10 hr)
  - Superior = $6,000 ($1,000/hr * 6 hr)
  - Premium = $6,000 ($1,500/hr * 4hr)
  - Total monetary cost of raw materials/day under new contract = $16,000

McKinsey & Company - 6

Case scope provided by the interviewer

A natural gas utility operates in 12 cities throughout 5 states. They are attempting to market a natural gas vehicles and appropriate filling stations. The utility company has spent $10M/year for the last 5 years. The CFO says an additional $50M (spread evenly over 5 years) is required. Specifically, the utility envisions turning diesel/gasoline fleet vehicles (UPS or Postal Service trucks for example) into natural gas vehicles. The CEO is a little skeptical of the CFO and wants us to determine if it is a good idea to continue.

Interview Hot Tip: The interviewer gave me absolutely zero time to think through a framework – he immediately asked me what my gut feeling was and started asking questions. You need to approach this problem from the viewpoint of the fleet owner – is it going to profitable for me to convert my vehicles? Search for applicable costs to each type of vehicle and compare.

Additional information provided after relevant questions

- State governments are pushing this conversion
- The natural gas market is growing (no % given)
- Customers will save $1/gallon after conversion
- Available fleet vehicles for conversion = 100,000 to 120,000 vehicles
- Conversion kit costs $2,500
- Pilot project on-going with 400 fleet vehicles – companies utilize both types of vehicles
  - Old Fleet Vehicle
    - 15,000 miles/yr
    - 10 miles/gallon
    - Diesel fuel = $1.50/gallon
  - Pilot
    - Fuel = $.50/gallon
- Client will make $450/converted vehicle.

Example - How to crack the case?

- One of my first responses was to tell him I was going to treat the $50M already spent as a sunk cost – the interviewer asked why and was not satisfied with the typical answer we learned from econ class. He kept asking me, "You don t think the CEO will have to answer to the Board and shareholders for a lost $50M". Don't back down, tell the interviewer that his statement is true, but for decision making it shouldn't be considered.

- Pilot project
  - Old Fleet Vehicle
    - 1,500 gallons of gas/year (15,000 / 10)
    - Total diesel fuel costs = $2,250/yr
  - Pilot
    - Total natural gas fuel costs = $750/yr (1,500 gallons * $.50/gallon)
  - $2,250 - $750 (saving from using natural gas) = $1,500
  - Year 1: -$1,000 Year 2: +$1,500 Year 3: +$1,500
- Client will make $45,000,000 if all vehicles convert, netting $35,000,000 per year.
- Numbers make it look attractive, assuming all 100,000 vehicles convert. In reality, less than 5% of vehicles will convert.

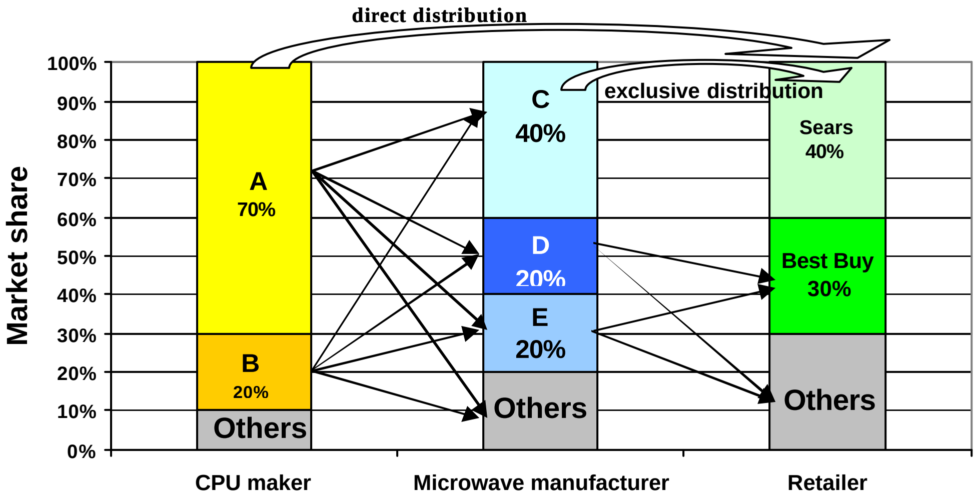
McKinsey & Company - 6

Case scope provided by the interviewer

A venture capitalist (VC) firm recently invested in the US microwave manufacturing industry by buying firm D (see chart below). Firm D makes microwaves by purchasing different components (most importantly, the Central Processing Unit or CPU of the microwave) and selling the finished product to retailers such as Best Buy. The venture capitalist thinks there is an opportunity to make CPUs in China.

Answer the following questions:

- Would you have invested in firm D if you were the venture capitalist?
- Should firm D integrate backward and relocate all or part of its production to China? If this is a good option, recommend the best way to do it.



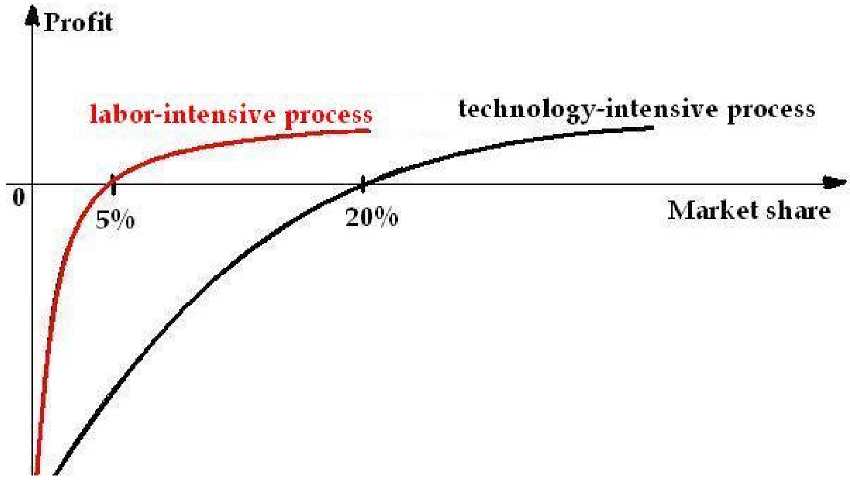
Additional information provided after relevant questions

- Clarifications regarding the graph:
  - Firm C has an exclusive agreement with Sears and sells only through them.
  - Firm A has begun selling CPUs directly to Sears to make a private label microwave.
- General data:
  - Currently all production takes place in the US (both by CPU and microwave makers).

- o Disregard transportation costs from China to the US.
  - Economics of the industry:
  - o The CPU is 80% of the cost of the microwave.
  - o Microwave manufacturers add little value to the product, they just do the final assembly.
  - o A is very profitable
  - o B is at break-even (0 profit).
  - o The microwave manufacturers (firm C, D, E) are all only marginally profitable.

Example - How to crack the case?

Key information needed for cracking the case: Since all production takes place in the US, where labor is very expensive, CPU makers currently use a technology-intensive process. The break-even volume for this process is 20% of the total number of units sold annually. However, a more labor-intensive process could be implemented in China, where labor costs are much lower, the break-even for this process would only be 5% market share.



Suggested framework:

1. **Economics** of the microwave industry
   a. Economies of scale
   b. Major costs
   c. Pricing
   d. Etc.
2. General **investment climate** in China

   a. Regulatory and political environment
   b. Economic conditions
   c. Necessity of local knowledge
   d. Etc.
3. **Costs** of relocating to China
   a. Investment required
   b. Incremental transportation costs
   c. Organizational impact
   d. Political risks, etc.
4. **Benefits** of relocation
   a. Cheaper labor costs
   b. Cheaper resources
   c. Etc.

Solution to the case:

- Since **microwave manufacturers** add little value in the manufacturing process, they will continue to have low bargaining power vis-à-vis the CPU manufacturers and especially the retailers, whose power is increasing. This is shown by the fact that some CPU makers are already going directly to the retailers. The decision of the VC to invest in firm D doesn't seem like a good one, because in such an environment, **profits will remain low**.
- Now let's look at the **CPU makers**. Currently, firm A is very profitable because with 70% of the market share, it produces much more than the break-even point for the tech-intensive process. Since firm B produces exactly the break-even volume it has 0 profits. The current technology-intensive process thus provided a huge **barrier to entry** into this industry. With the new labor-intensive process, new entrants can enter the market much more easily and make profits even at low volumes.
- The future of **firm A** looks bleak, because a price war is inevitable as new entrants break into the market and try to achieve the 5% break-even volume. This will drive down profits for everyone and a fragmentation of the industry will occur. Given the economics of the industry, the **share price** of firm A **can only go down** from this point as profits decline.
- The long-term future of **firm B** is similar to firm A. However, in the short run, it can relocate production to China and make a profit even with 20% of the market share. It should leave prices unchanged in order not to start a price war with A who has a cost advantage. If firm B can execute this strategy, its **share price may rise in the short run** as it begins to make a profit. In the long run, however, firm B's profits will also be competed away by new entrants. The share price will fall as profits also decline and competition increases.

- The **venture capitalist** should **exit the microwave industry** in the long term. A **short-term investment in firm B** makes sense only if we assume that the company would have a few years' lead over other firms in relocating production to China. If new entrants will not copy B's strategy immediately, then B can make profits for a few years by relocating production to China. The firm's share price will rise in the short run and the VC will make money with this investment.

AT Kearney - 1st Round - 1

Case scope provided by the interviewer

Agricultural goods manufacturer is having issues with compact tractor division (division sells tractors in Southeast U.S.). Inventory is growing (both raw materials and finished goods), delivery performance to retailers is getting worse, and too much money is being spent on purchased parts which often aren't the parts needed. The CEO would like our help in diagnosing and solving the problem.

Additional information provided after relevant questions

- One supplier (Japanese) provides 70% of the parts for compact tractors
- Requires 60 day lead time
- Commercial retailer requires 4 week lead time
- Unstable demand
- Consumer has choice between automatic and manual
- Tractors shipped via truck
- 100 tractors can be built/day with proper parts

Example - How to crack the case?

Perfect example of how an interviewer will walk you through a case if you structure the problem correctly. I drew a framework concentrating on

1) Supplier 2) Cash position and 3) Commercial suppliers. He told me that another AT Kearney team was looking at the cash position and commercial suppliers so we could concentrate solely on suppliers.

- Points to consider:
  - Seasonality of demand for both retailer and supplier
  - Is lead-time really 60 days?
  - Examine historical data to get a feel for monthly demand
- Japanese supplier should incorporate buffer inventory
- Supplier should purchase warehouse in U.S. to store parts that client cannot use. Client should not pay for parts until delivery to manufacturing plant.

AT Kearney 1st Round - 2

Case scope provided by the interviewer

Our client is a European auto-parts manufacturer that supplies any Auto company you can think of (GM, Ford, VW, Honda, etc). They are also present in the auto after-market, that is to say they sell some branded goods in the retail market in Europe.

Our client wants to enter the U.S. retail market. They have lots of other branded products, but they asked our help specifically for their windshield wiper product. Make a back-of-the envelope market sizing and then come up with a competitive strategy for this product.

Conversational example - How to crack the case?

There are almost 300M people living in US, and 100M households. If we assume that every household has one car in average, we can say that there are 100M cars on the roads in the US. Windshield wiper is a replacement product; the new cars already have it and do not need it. The old cars need to replace their wipers once a year. That means the market for the windshield wipers is 100M units/year.
Yes, our estimation was exactly the same with yours. Average price in the market is $3.

Who are the competitors in this market and what is their market share?
There are three main competitors: Anco (38%) and Trico (31%) are the traditional players, Bosch has come to the market 3 years ago and captured 14% market share.

What has Bosch done differently in order to capture this market share?
There was only standard product in the market. Bosch has introduced a premium product. But, they had to spend huge marketing $s. We can compare their spending as 10 to 1 with their competitors.

What product segments does our client offer in Europe? What is our client associated with? Quality or low-price?
Our client offers both premium and standard products in Europe and it has a very strong brand equity associated with quality in Europe after-market.

What channels are used to reach the end-consumers and what are the breakdown %s?

	Volume	Value
Auto-parts retailers	30%	40%
Wholesalers	45%	40%
Franchises (like gas stations)	5%	5%
Mass Merchandisers	20%	15%

AT Kearney 1st Round- 3

What would you consider if you where to evaluate source auto components from China?

I considered. Incremental profit (labor, material?, low capital investment, etc.), Effect on current operations (such as will sourcing from China support JIT?, materials available, communications both verbal, IT, ordering systems), Effect on Current Supplier's (may create confirm if choosing different suppliers, etc.)

How would I determine which products to source from China?

I considered Labor Intensity, Volume, Length of program, Complexity of product, Ease and cost of shipping, and Possible Government Regulation.

Part C: How would you transfer manufacturing of existing components to China?

Create a project team and needed infrastructure, Evaluate options for transfer (Use Chinese supplier, request International supplier to local in China, or Direct Investment), Determine objectives, Create a pilot program (transfer simplest product/process first), Measure results compared to objectives, Scale up project to include transferring the other projects.

AT Kearney 1st Round- 4

Case scope provided by the interviewer

My client, the CEO of a large chemicals manufacturer, is in the process of evaluating a merger opportunity with a competitor. He calls me at midnight and requests me to help him justify a $200 million benefit (including one time and recurring) from this merger. There is limited time as he needs to make an announcement early next morning to the Wall Street Analysts covering his company.

Additional information provided after relevant questions

- Synergies between two companies in the area of chemicals being manufactured for polymers
- 3 players account for 90% of market share in the above business
    - Company A (client company) has 30% market share
    - Company B (the company being merged) has 20% market share
    - Company C (competitor, part of a diversified chemical conglomerate) has 40% market share
- Product range (of polymer business) of three companies:
    - Company A: Mid and low-range products with moderate margins
    - Company B : Premium product with high margin
    - Company C: Products spanning all three (premium, mid and low range)
- Client company (Company A):
    - Revenues - $2 billion (80% from chemicals used in making polymers, 10% from chemicals used in Agrochemicals, 10% from chemicals used in additives)
    - Variable costs (Major cost areas: Raw material – Petrochemicals, Packing Material, Selling and Distribution costs, operating and utility costs for plants)
    - 86 manufacturing plants and 45 distribution warehouses all over the world
- Company B
    - Revenues - $1 billion (all from chemicals used in polymers)
    - Variable costs (Major cost areas: Raw material – Petrochemicals, Packing Material, Selling and Distribution costs, operating and utility costs for plants)
    - 35 manufacturing plants and 20 distribution warehouses all over the world
- Supply-Demand
    - Demand outstripping supply

- Share of mid-range ad premium products on the rise vis-à-vis low range products

Example - How to crack the case?

Framework & Structure:

This is a merger case. The emphasis was not on arriving at the exact number, but on the approach taken to identify key opportunity areas. Once I identified each opportunity area and attempted to size it, the interviewer provided me the benefit figure.

- I used the 3C framework to first understand the market (customers and competitors)and then the positioning of the two companies in this market.
- I then used the Profitability framework to understand the key drivers of Revenues and Costs for both companies.
- Based on the initial 3C analysis and the Profitability framework, I attempted to zero in on major opportunity areas.
- I then sized each one of them to arrive at the $200 million (approx.) benefit figure.

Opportunities identified

- Revenue increase
  - Opportunity to cross-sell products
  - Opportunity to up-sell products (convert customers of midrange product to premium products)
  - Raise prices due to greater market power
- Cost reduction
  - Asset optimization
    - Reducing number of manufacturing plants and number of distribution warehouses
    - Reducing the number of sales offices
  - Network optimization - Redesigning the distribution structure to ensuring lowest cost of delivery; reduction in transportation costs
  - Procurement costs reduction
    - Significant reduction in packing material costs with increased negotiating power (packing material industry is highly fragmented)
    - Limited opportunities in raw material (petrochemicals) costs as they is a commodities and increase in purchasing volumes will not result in substantial savings
  - Headcount reduction:
    - Largely in sales, procurement and operations

Key aspects of the case that the interviewer seemed interested in testing me on:

1. Methodology used to hit all big opportunity areas (the 3G-Profitability framework seemed to work well with the interviewer)
2. Approach for sizing big opportunities - Cross sell/ Up sell, Asset optimization and Network optimization
3. Identifying the risks of merger – Culture clash, attrition of employees, loss of existing customers

AT Kearney 1st Round- 5

Case scope provided by the interviewer

Your client is the largest supermarket chain in Bolivia. The CEO of the company has noticed that barcode scanners are becoming a major trend in the European and North American market. He wants to know he can and whether he should install them in his own stores.

Hint:

You can get two types of case questions. One will be long and crammed with data. In this case, a part of the data may be there to confuse you and help the interviewer evaluate whether you can select the most relevant data. The other type, like this one, will be very brief, 2-3 sentences. In a brief case question like this, every word typically matters. Be sure to listen carefully and jot down all relevant phrases.

Additional information provided after relevant questions

Feasibility:

The technology required for barcode scanners is a mature technology available from several suppliers. IBM, HP and others would be willing to supply the Bolivian market.

Assume that putting the barcodes on the products would not be a problem.

Current system vs. barcodes:

Printed book of prices contains prices for each product. Based on this book, dedicated shop assistants place price stickers on all products throughout the store. The person at the cash register then types in the department code in the cash register (which is not connected to any server right now) and the price based on the price sticker (the people at the cash register also have the book of prices with them should they need it). Then, at the end of each day, a central research system collects the sales data from each cash register and gathers it into a central MIS. This information is used to manage inventory.

With barcodes, all cash registers would have a point-of-sale system that would be connected real-time to a central server.

Competitors:

70% of the market is controlled by street vendors.
30% supermarkets, but most of them are stand-alone, not chains. The client controls 6% of the market, while the next biggest chain only holds 0.6% of the market.
Competition is pretty intense, because there is always a small store a street vendor on the corner who competes directly with the supermarket chain, mainly on price.

Costs of system:

$1 million/store fixed cost
60 stores
+10% of initial investment is an annual maintenance cost

This includes everything, from training to IT, etc.

Hint for case-giver:
The interviewer didn't wait for me to make all the calculations, he just wanted me to conclude that it's a damn big investment and we are going to have to find some huge benefits to offset this.

Benefits of the system:

Labor cost savings:
The supermarket chain carries some 6000 SKUs (stock-keeping units). It employs 10 people dedicated to placing price tags on products in each store. They are paid 4 dollars a day. There would be no need for them if we had barcodes.

Better service:
The average time spent in the store is 30 minutes. Right now people spend 3 minutes on average waiting in the check-out line. They spend 1.5 minutes waiting and 30 seconds paying. The remaining 1 minute is the time taken by the cashier to enter the prices of the products in the cash register. This would be cut down to 6 seconds under the new system.

Economic environment:

Key to the case: what is the inflation rate in Bolivia? It is huge! Therefore, the stores must reprice everything constantly. That is why they employ 10 people per store just for this, but even so, they cannot respond quickly enough.

Each store makes $20 million/year and they have a 3% profit margin.

Example - How to crack the case?

Proposed framework:

Since the interviewer asked two questions, you should separate the framework into 2 parts: feasibility and a cost/benefit analysis.

Framework

1. **Feasibility**
    a. **Availability**
    b. **Stores**
        i. Location
        ii. Available infrastructure
        iii. Feasibility of training (work force abilities)
    c. **Competitors**
        i. Benchmarking
2. **Cost-Benefit Analysis**
    a. Understand the Bolivian **environment**
        i. Economic environment
            o Inflation
            o Labor costs
        ii. Regulatory environment
        iii. Political environment
    b. **Benefits**
        i. Tracking orders
        ii. Improved service
        iii. Cost savings
        iv. More accurate product pricing
    c. **Costs**
        i. Fixed cost implication
            o IT investment
            o Training cost
        ii. Variable cost
            o Labor
            o Materials
3. **Strategic Implications**

Solution:

Assuming a 10% dollar interest rate, the net present cost of implementing the barcode system in all 60 stores is $60 million+6/0.1=$120 million.

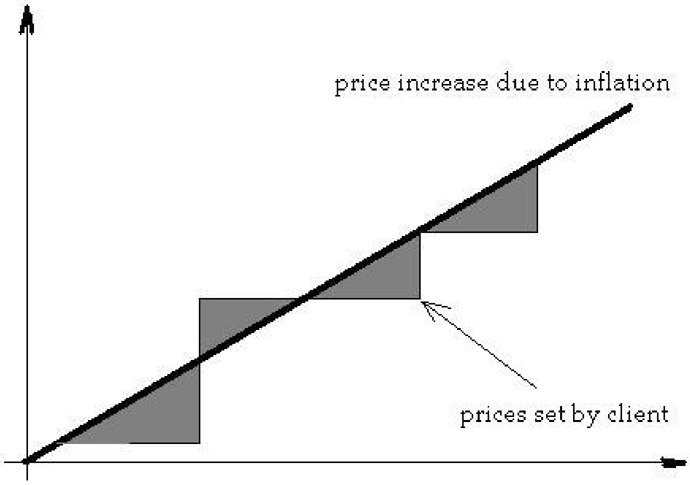
I have found quantifiable cost savings of ($4 x 10 x 60 x 365)/0.1 = $8.76 million or roughly $10 million.

The reduction in check-out time by 54 seconds is not likely to make a significant difference to customers, since they spend 30 minutes in the store and the basis of competition is mostly price, not service quality.

These 2 changes would, in themselves, not justify the new barcode system. However, the fact that inflation is so huge in Bolivia is an interesting fact. As stated before, the stores have a hard time keeping up with price changes. Since competition is fierce and is based on price, the stores can maximize profits by responding at exactly the same time as their competitors, not a moment before or after.

This below graph illustrates this idea:

The straight line represents the general increase in prices at food stores due to inflation. However, the supermarket is not able to increase its prices continuously under the current system. Each time they raise prices later than the competition, they miss out on profits and each time they raise prices sooner than the competition, shoppers go elsewhere, so they lose sales. The shaded areas in the graph, thus represent their lost profits. This is the main benefit that can be derived from the new barcode system. We now that the supermarket chain makes 20 x 60 = $1200 million in sales 1200 x 0.03 = 36 million in profits. Assuming that inflation will remain high for the foreseeable future, the barcode system should be implemented if faster reaction to price changes can increase margins by at least 1%. (110 million x 0.1 = 11 million = cc. 1% profit margin)



Hint:

9 people out of 10 probably will not get this last point. In my interview, the interviewer clearly thought that this was the "key to cracking the case", but he didn't think it was the key to doing well on the case. I needed a lot of help from him to come up with this solution he even started drawing the graph for me, but he still passed me on to the next round because I was well structured and consistent in my approach. Lesson: Staying calm, not getting frustrated when you miss a key detail and keeping a good structure are much-much more important than cracking the case. Remember, they are also evaluating you "teachability" (your listening skills and how easily you catch on to new ideas), interest in cases, overall attitude and thought process, not only raw smarts.

AT Kearney 1st Round- 6

Case scope provided by the interviewer

I work in a major office building. I noticed that there were not any coffee stands in the building and that everyone had to walk one block to the local Starbucks to get coffee. I thought that I could purchase the rights to sell coffee in the lobby, hire someone to run the booth and make some income in addition to my salary as a consultant. Tell me how much I can make in one year.

Interview Hot Tips: This case is a profitability case. You are going to be making up the numbers so make sure you assume easy numbers to calculate so that you don't fumble the analytics of the case.

Conversational example - How to crack the case?

I want to address the issue of whether to open the coffee stand by looking at the profitability framework. If we find that at the end of the analysis, that the total revenue is greater than the costs of opening the coffee stand, then you should definitely start this business to earn ancillary income. I will also let you know, based on some numerical assumptions, how much you should expect to make in one year.
Great. Let's get started.

First, I want to look at the cost side. How much would a permit cost for one year to operate in the lobby of your office?
$2000

Does this license provide you with exclusive rights to sell coffee in the lobby or can competitors enter the market?
This is an exclusive license. Others can come in and sell different products but I will be the only coffee vendor.

Great. Now I would like to look at the fixed cost side. How much does a coffee cart, coffee maker and the basic necessities needed to brew coffee cost?
Let's say the total initial investment is $5000 for the designer cart, stool, coffee maker, power supply, aprons, awning, marketing sign stand, music speakers and cd player, etc.

Okay, now I would like to look at the variable costs. For each coffee sold, we need to provide a cup, napkin, lid, and coffee cup sleeve. On average, how much does this cost per cup of coffee sold?
Since we will be buying in bulk, we can get the variable cost down to 35 cents per cup of coffee sold.

I will come back to that number once I size the market. However, I still need to round out the cost side. I would also like to look at the amount of money you are going to have to pay an employee on a yearly basis. I would figure that you will be open 250 days a year for 8 hours a day. That is a total of 2000 hours. On average you will pay your employee $15 per hour for a total of $30,000. That $15 dollars includes any bonuses you may pay to the employee. Let's also include $500 for training of each employee. Since we will have 2 employees – we will need to spend $1000 on training. So for employee costs we will spend $31,000 per year.
Great.

Now let's move on to the revenue side. How many people work in the office building?
2500

Okay. So at least 2500 people enter the lobby on a daily basis. But, we need to augment that number to reflect the other people that come into the lobby – like clients who come to visit the companies within this office building and others. So let's say 3000 people enter the lobby on a daily basis.
That is a good assumption.

Out of those 3000 people, let's assume 30 percent purchase a coffee from your cart. So we have 900 coffee sales per day. What are going to be our product offerings?
Well, for the sake of simplicity, let's say the average purchase is $1.20

Great. Each day, on average, the revenue is $1080. On a yearly basis, given the fact we are open 250 days a year, our total revenue will be $270,000.
Sounds like a nice stream of revenue.

So now let's go back to the cost side and figure out the variable costs based on our market size. If each purchase costs us 35 cents, then we take 900 times 35 cents times the 250 days per year we are open and we get and annualized variable cost of $56,250.

Now let's look at the profitability equation:
Total Revenue: $270,000

Total Cost: $78,750 (for the variable cost) + $ 31,000 (for the employee cost) + $5000 (for the initial set up cost) + $2000 (for permit)

Therefore, we are making a profit of $153,250 on a yearly basis. This looks like a nice business to enter.

Thanks for solving the case.

AT Kearney 1st Round- 7

Case scope provided by the interviewer

A storm door company has received a new contract from a large home improvement chain (such as Lowe's, Home Depot, etc). The contract offers the company to sell all its products to the chain stores. The contract will double the current production and it must be delivered one year from the date of contract. The chain will buy the products with the current price, but in 4th year, it asks for 5% discount per year afterward. The contract lasts for 5 years and can be renewed after that. The storm door company can not sell its products to any chain competitors and can not communicate any issue with its competitors or buyers for the first 2 years. If the storm door rejects the contract, the chain will shut its supply from the storm door company. The sales to chain store accounts for 30% of the total production. The company came to you and asked for suggestion whether to take the offer or not.

Example - How to crack the case?

It was a Porter's 5 forces classic case (I didn't get it though). It's about bargaining positioning between buyer and supplier. By taking the contract, the storm company will very much depend on the chain. With the profitability will decline after 4th year, the company will certainly lose money. At the same time, if they don't take the offer, they will lose 30% of its production.

Take the contract and after two years (and profitable) starts talking to competitors or companies that sell products to this chain store and have similar contract. With the profitability, the company was then sold to its competitor that sells kitchen cabinets to the chain store. The merger creates higher bargaining power, which will enable company to make money after the contract is over (after 5 year).

AT Kearney 1st Round- 8

Case scope provided by the interviewer

A regional electric and gas utility has high costs in it customer service centers. What is the problem and how can it be resolved?

Conversational example - How to crack the case?

I would like to look at three areas: How high are their costs?, Why are the costs high?, and What can be done to reduce the costs? I would first like to compare their costs to their historical performance and then to similar utility companies. Has the cost of the customer service centers been increasing as a percentage of sales?

The current cost have been similar to the companies cost in the past

So this has not been a recent development due to a shortage of gas or due to problem with a new billing system. It is a system problem. Is this u tility a local monopoly?

Yes

Are their any comparable utilities that we know of? Where we could compare the customer service costs?

There are other utilities that are comparably. They spend about 20% less on customer service.

Labor rates, lease rates, taxes will be different from region to region. Have these been taken into account?

You can assume so for this case

It seems that we have a systemic issue resulting in a 20% higher customer service cost relative to comparably utilities. I would now like to look at why the costs are high. Does this firm's call centers handle a higher volume of call per customer than other utilities. Perhaps the problem is our gas or electric line are unreliable, or our billing system constantly generating errors.

No the volume of calls is in line with other firms.

What is the cost structure for customer service? I expect that there are a number of fixed costs such as buildings or leases, equipment. Variable costs of telephone charges. Labor may be a combination of fixed or variable.

There are our main costs; Buildings/leases at $2M per year, Technology $.75M, Telephone charges of $1M, and labor of $5M

Are these cost similar to our competitors on a per customer basis?
Buildings, technology, and telephone charges are similar but our labor costs are higher

Since our labor costs are higher I would like to look at labor productivity, labor rates, capacity utilization.
There are three types of employees and you can ignore capacity utilization:
Union Employees – Handle 20 calls per hour, cost $15 per hour.
Type 1 – Handle 30 calls per hour, $7.5 per hour.
Type 2 – Handle 25 calls per hour, cost $10 per hour.

How many employees of each type work for this firm?
Each type is about 1/3

What about their rates of handling customer service issues during the first call.
Union employees handle the issue during the first call 100% of the time, type 1 50%, and type 2 100%.

The "true cost" per call for the different types of employees would be:
  - Union - $15/20/100%=$.75
  - Type 1 - $7.5/30/50%= $.50
  - Type 2 - $10/25/100%=$.40

Can you tell me about the different types of employees:
Union employees are just that. There are certain sites that are all or mostly union.
Type 1 employees are new employees who have been with the firm for only 1 or 2 years.
Type 2 employees were type 1s but now have more experience and have been with the firm more than 2 years.

Do the other utilities have similar types of employees?
Other companies have more of the type two employees and fewer union employees.

From our investigation it appears that the utility is suffering a cost disadvantage due higher labor costs. I would like to investigate ways to lower these costs.
- Potential ways to lower costs:
  - Retain more Type 2 and also Type 1 employees who will become type 2s. This could be done by:
    - Shifting work from unionized call centers.
  - Incentivize employees to increase productivity. This could be tied to a measure of calls per hour and number and number of repeat calls.

- Provide training to type 1 employees to increase their ability to handle calls on the first call.

AT Kearney 1st Round- 9

Consider a typical paper mill. What do you expect are the inputs for paper production?

Pulp, some type of binding agent to glue the pulp together (Slurry), bleach, dyes, tools, presses and other equipment.

What processing steps would you expect to see in a mill?

- Mixing – to mix the paper and slurry together.
- Rolling or pressing – to remove water and flatten the mixture.
- Cutting – Cutting the paper to the cor rect size.
- Dying or bleaching – To create the correct color.

Where do you think the paper goes after leaving the mill?

- Distributors
- Direct to retailers
- Direct to customers

Why do you expect some customers purchase direct and others through distribution?

- Size of the company and the amount of paper they purchase. Larger companies who buy large sums can often purchase direct and save a large amount of money.
- Type of product – Companies may require special dyes or types of paper, which led them to buying direct/

AT Kearney (2nd Round) - 1

Note: *You will only get this type of assignment with AT Kearney and only in 2nd rounds!*
Instructions: *You have 60 minutes to read the case and prepare the slides and are expected to hold an approximately 15 minute presentation followed by 5 minutes of questions. You may use a calculator, markers and transparencies.*

Medical supply chain

As you left the meeting with Tom Owens, VP of Logistics, you were confident that you would be able to shed some light on the supply chain issue of Medicine One. Hugh Martin, CEO of Medicine One gave Tom the green light to establish the distribution network for the entry to the European market after seeing the promising sales projections from Marketing. (Table 1)

The Company

Medicine One is a worldwide developer and manufacturer of less-invasive medical devices. Its products are used to diagnose and treat conditions in a wide variety of medical fields, including cardiology, gynecology, oncology, radiology, urology and vascular surgery. Products fashioned by Medicine One include catheters, surgical grafts, coronary and ureteral stents, polypectomy snares and lithotripsy devices. The Company's products are offered by two dedicated business groups, Cardiovascular and Endosurgery.

The Company's history began in the late 1970s, when founder Hugh Martin acquired an equity interest in Medscope, an R&D company focused on developing alternatives to traditional surgery. In 1985, Hugh and Kate Porter partnered to buy Medscope and together formed Medicine One. Driven by the needs of the medical community for innovative products, the Company grew from revenues of $1 million in 1985 to more than $500 million in 2001.

The medical devices industry has been faced with low growth opportunity over the last five years. There are a lot of margin pressures due from the HMO and the consolidation of hospitals.

The Problem

Medicine One currently has manufacturing in Lake Forest, IL and Raleigh, NC and serves the US market. The plant in Raleigh has ample capacity for the projected sales in

Europe. You have to look at the most appropriate distribution method for the European sales. The VP of Logistics is considering two options.

The first option is to have a one-hub distribution center (DC) in Berlin, Germany and 20 satellite warehouses in relevant European cities that directly serve customers. Initially inventory levels will be set at 100 days in the Berlin DC and 30 days in each satellite warehouse to meet projected customer demand.

The second option is to have 4 DCs in Berlin, Germany; Paris, France; Gothenburg, Sweden and Milan, Italy with no satellite warehouses. Berlin DC will serve the demand in Polish cities and the Paris DC will serve the demand in English cities. Initially inventory levels will be set at 150 days in each of the DCs to meet demand for countries and cities served.

The high inventory level is necessary to support the aggressive sales targets in the first year but may not be sustainable long-term.

In the US, Medicine One has next-day delivery to its customers and wants to establish similar criteria in Europe, if feasible. Tom is worried about the transportation costs and the amount of capital required to operate in Europe. Tom indicated that services levels and cost are the key criteria in determining how to set up the European Supply Chain. Service level is dependent upon the % of demand that could be delivered the next day, 2nd day, 3rd day, etc. And the total cost is composed of transportation, inventory, warehouse and capital costs.

Transportation cost

All inbound transportation is by air and would take an average of three days from NC. However, there is no cost difference among flying to different cities in Europe. The distribution between countries would be through ground transportation. For option one, the transportation costs from the Berlin DC to the satellite warehouses in each country are shown in table 2. In option one deliveries from satellite warehouses to the customers and in option two deliveries from each DC to the customers would cost $10/pkg.

In option one the shipment is delivered to the customer one day after it has been shipped out from the local warehouse. In the second option, it takes up to three days to deliver the shipments from the distribution centers to the customers according to the distance served. The service level for option one is 100% next day delivery. However, with option two 50% is next day, 30% is 2nd day and 20% is third day delivery.

DC and Satellite Warehouse Cost

DC and satellite warehouse operating costs are on average $30 per square meter including lease, utilities, labor and depreciation. The capital cost for a DC is on average $60 per square meter including building, equipment and initial shipment of inventory (not safety stock investment). Medicine One wants to own the DCs and lease public facilities for the satellite warehouses. You can ignore the markup for public warehouse services. The minimum size for a distribution center is 1000 m^2 and space requirements are calculated at 0.1 m^2 per unit of package.

Inventory

The annual inventory carrying cost is 20% and the actual cost of the inventory is 40% of the sales price. Medicine One operates 300 days a year. The costs of initial inventory levels are included in the capital cost above. The amount of capital invested in inventory will significantly increase Medicine One's working capital on the balance sheet.

Your Presentation

Tom has asked you to advise him in establishing a supply chain structure for the European Market. He wants you to come up with the most cost effective option that meets customer service requirements and future demand as well as help him think through the decision process.

1. Develop an analytical plan to identify the capital requirements and operating costs (1 slide approx.)
2. Analyze the two different supply chain configuration options and a provide recommendation (detailed calculation)
3. Identify key supply chain issues and potential trade-offs in this distribution problem that will help the client think through the decision. (1 slide approx.)
4. Outline a better solution than the options presented and also, discuss what else should be taken into consideration that would lead to lower costs and/or improved service levels. (1 slide approx.)

You need to make assumptions as you develop your analysis, be prepared to discuss what they are and the logic behind them.

Note:

Ignore any governmental regulation issues that may be pertinent to medical devices.

Table 1: 2003 European Sale Estimates

Country	Package Volumes (000 pkg)	2003 Sale Projections ($ in MM)
Germany	20	20
France	15	15
England	5	5
Sweden	10	10
Poland	4	4
Italy	12	12

Table 2: Average Pan-European Transportation Costs for the 1 DC Model

Destination country	From Berlin DC ($/pkg)
Germany	8.5
France	9
England	9.5
Sweden	9.5
Poland	9
Italy	10

Sample presentation:

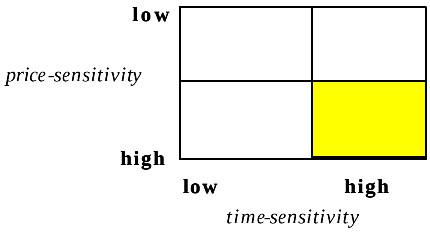
Agenda

1) Recommendations *(NOTE: ATK usually likes recommendations first!)*
2) Cost evaluation (option 1, option 2)- comparison
3) Analysis of tradeoffs
4) Alternatives

Slide 1

Recommendations

1) Chose option 1 as a platform to improve upon.
   - Slightly higher annual cost, but guarantees higher customer satisfaction.
2) Target Time -sensitive high-end price segment



3) Evaluate European environment to implement optimal supply chain strategy
   - Customers, competitors, regulations, etc.
   - Future trends (aging population, but price sensitive healthcare, private clinics)

Cost evaluation

Framework used for the financial analysis
(factors to be considered)

Capital requirements

1. Investments in DCs
   - minimum size: 1000 m^2
   - space requirements: 0.1 m^2 /pkg
   - owned by Medicine One
   - cost/m^2 = $60
2. Investment in inventory
   - initial inventory

Operating costs

1. Transportation
   - Shipping cost to Europe (common for option 1 and 2)
   - Shipping cost to warehouses (only option 1)
   - Shipping cost to customers (same for both options=$10 x 66000 units=$660K)
2. Inventory-holding costs
   - COGS+ inventory carrying cost= 60% of the sales price
3. Warehouses
   - leasing in option 1

Option 1

Capital requirements:

1) DC in Berlin
   - Annual European sales = 66000 units
   - 100 day inventory= 22000 units
   - 0,1 m^2/ pkg ⇒ 2200 m^2 needed
   - \$60/m^2 cost for DC

   = \$132 000

2) Initial inventory
   - Average sales price= \$66 MM / 66000 pkg = \$1000/ pkg
   - 1 Berlin DC + 20 whouses = \$600 x 22 000 + \$600 x 30 x 220 = \$17.16 million
   - significant increase in W/C (included in Capital costs)

Operating costs:

1) Warehouse leasing
   - 30 days inventory= 6600 units
   - 0,1 m^2/ pkg= 660m^2 needed

   at minimum (optimizing for distribution time may require higher levels)
   - \$30/m^2 =

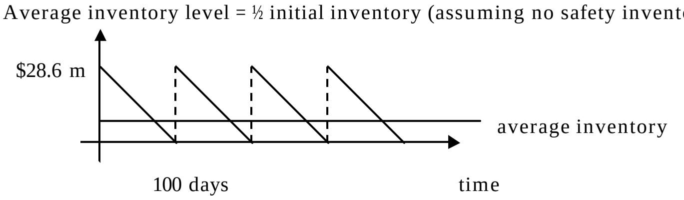
   \$19800/ year leasing costs

2)Transportation Costs

	Volume	Cost/pkg	Total cost
Germany	20	8.5	170
France	15	9	135
UK	5	9.5	47.5
Sweden	10	9.5	95
Poland	4	9	36
Italy	12	10	120
			\$603500/year

3)Inventory carrying cost

Average inventory level = ½ initial inventory (assuming no safety inventory) = 28600 pkg



Inventory carrying cost = \$28.6 / 2 * 20% = \$2 860 000

Total cost of option 1

= \$132 000 initial cost

+ \$19 800 + \$603 500 + \$2 860 000 annual cost = \$3 483 300/year

Service level

100% next day delivery

Option 2

Capital requirements:

1)Investment in DCs

- 4 DCs
- 150 day inventory = 33 000 units
- 3300 m^2 needed
- $60/$m^2$

= $198 000

2)Initial inventory level

- 60% x 33 000 = $19.8 million

Operating costs:

1) Warehouse costs - none

2) Transportation costs

	Volume	Cost/pkg	Total cost
Poland	9	12	108
UK	9 (assumption!)	5	45
			$153 000/year

3)Inventory carrying costs

- 33 000 initial inventory level
- 33 million / 2 x 20% = $3 300 000

Total Costs of Option 2

= $198 000 initial cost

+ $153 000 + $3 300 000 annual cost = $3 453 000/year

Service level

50% next day, 30% 2nd day, 20% 3rd daydelivery

Summary slide – Option 1 and 2

	Capital investment (without investment in inv.)	Annual cost	Service level
Option 1	$132,000	$3,483,300	100% next day
Option 2	$198,000	$3,453,000	50% next day

Conclusions:

- Option 2 saves $30 000/year in operating costs (small cost difference!), making up for the higher initial investment in 2 years.
- BUT: Option 2 offers a significantly lower service level.
- Thus, if the European market is of strategic long-term importance and customers are more time-sensitive than price sensitive, chose option 1.

Key supply chain issues and potential trade-offs

Issues to consider in the supply chain decision:

1. Inventory levels
   i. Customer service levels (option 1 outperforms option 2)
   ii. Holding costs of inventory + warehouse and DC size
2. Capacity and production
   i. Local production ⇒ less transport cost, better access to markets
   ii. Export ⇒ use available excess capacity
3. Transportation
   i. Outsourcing (is it possible?)
      - Control issues, time-sensitivity (reliability of partner)
   ii. Own fleet for transporting
      - Capital investment, management know-how?
4. Fluctuations demand (variability leads to higher costs)
   i. Ordering and planning/scheduling systems
   ii. Safety capacity in warehouses and DCs

Key Trade-offs

Aggressive growth vs. High-quality service vs. Cost vs. Timeliness

Recommendations:

Meet aggressive growth targets by

1) examine possibility of local production in the medium to long-term
2) ensure appropriate customer service level
3) ensure reliability of transporters

Alternative solution

- Manufacture products in Germany
- outsource (shorter term)

build facility (longer term

Issues

1. Quality issues
   - quality standards
   - licensing – risk of technology appropriation
2. Governments/regulations
   - lengthy/costly approval process for new facility
   - higher volumes needed to spread fixed cost – long-term commitment to European market
   - better ability to respond to changes in pricing, demand
   - local manufacturers preferred in some markets
3. Customers
   - Issue of access to customers
   - Segments to be targeted
     - Public hospitals (large vs. small) or Private clinics
   - New markets (e.g. Eastern Europe)
4. Competitors
   - New entrants
   - Established players

AT Kearney (2nd Round) - 2

Case scope provided by the interviewer

Your client, an automotive OEM, wants to understand the dynamics of the remanufactured parts market. In order to better help him, estimate the annual size of the remanufactured transmission market in the US.

Facts:

- Total remanufactured market = $10 billion annually
- Power-train sub-segment is about 50% of that:
  - Engines
  - Transmissions
- Average life of a transmission = 7 years
- Average life of a car = 17 years
- Value chain:
  - Two types:
    - Dealer – Remanufacturer– Distributors– Customers
    - Independents – Remanufacturer– Distributors– Customers

Conversational example - How to crack the case?

Before I explain how I will approach this problem, I would like to make sure I understand the problem. Can you tell me more about the transmission remanufacturing process and how the OEM is involved in this?
Sure. When a complex auto part is broken, it is usually only one of the inside components that has to be replaced. When a transmission breaks, the valuable outside hull can still be reused, it is only the inside the must be changed. This part is then taken and resold. The OEMs usually outsource this remanufacturing activity.

Ok. I want to approach this issue from two directions. First, I will use numbers regarding the total remanufactured market value to arrive at an estimate for remanufactured transmissions. Second, I will figure out how many transmissions break annually based on an estimate of the total car pool in the US. Finally, I will compare and contrast these two approaches and explore reasons for any discrepancies between the two estimates.

Sounds good. Actually a third approach you might take is just to use expert estimates, but clearly you can't just take those at face value unless you know how they were computed. So, where do you want to start?

The total remanufactured market is $10 billion. 50% of that, or $5 billion is power-train, which consists of engines and transmissions. If I had information about the price of the two parts and their average life, I could estimate the share of transmissions from this $5 billion.

So how will these two factors affect the size of the market?

I expect transmissions to have shorter expected lifespan, meaning that they should have a larger share of the market. However, a transmission costs less than an engine, so based on price, I would expect transmissions to have a lower share.

So based on this, what would be your guesstimate?

If we do not have more precise information, it would be reasonable to assume that engines and transmissions have an even share of the market in terms of value, since transmissions brake more often than engines. Thus, the value of the remanufactured transmissions market is $2.5 billion.

Is that where you would stop?

I would also like to know the volume of the transmission market to be able to compare it with the figure I will derive from my second approach. Do we know the average price of a remanufactured engine?

These parts sell for about $1000 in the store.

But we are trying to estimate the size of the market for an OEM, who would be interested in the total revenue for re-manufacturers. Clearly, distributors make some margins, so re-manufacturers must be selling these parts for less than $1000.

Yes, distributors have a 25% mark-up.

Ok, so the manufacturer's price is $1000 / 1.25 = $800. This means that the volume of the transmissions market is $2.5 billion / $800 or roughly 3 million transmissions per year. This doesn't seem like a very large number. Let me try to confirm this number using a different approach.

Ok.

Let's assume there are 100 million households in the US, each with 2 cars/household. Is that about right?

It may be a little more than 2 cars/household, but let's go with 2 for now.

So that is 200 million cars in the US. We know that a transmission breaks down every 7 years on average, so the probability of break-down in any given year is 1/7. This means that about 30 million transmissions break. But that means only about 10% of broken transmissions are replaced through remanufacturing.

What might be some of the reasons for that?

If the $10 billion figure includes all remanufactured parts, then there may be several reasons for this discrepancy. Some people will buy totally new parts, maybe because they are worried about the quality of remanufactured parts. Others may decide not to replace a broken part because it would cost more than the entire scrap value of their vehicle. This is particularly true if they have an older car that has multiple problems or broken parts. For some rare models, remanufactured parts may not even be available.

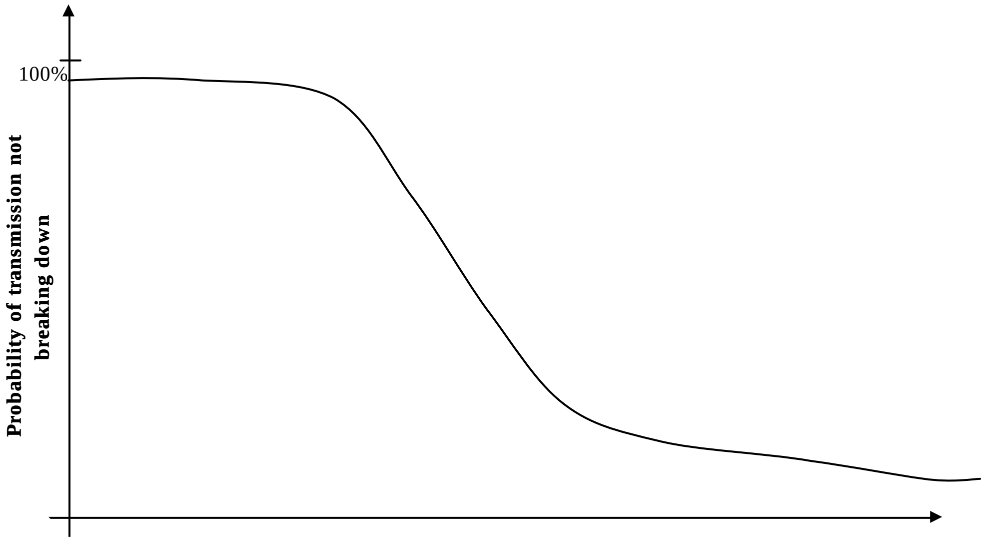
Ok, let's suppose that you came across an expert estimate that suggests that 25% of broken transmissions are being replaced through remanufacturing. Assuming that 3 million remanufactured transmissions are being sold annually, could that be an accurate figure?

That would mean there are only 3 / .25 = 12 million transmissions breaking annually. This is considerably less than the 30 million estimate I arrived at. Assuming there are 200 million vehicles in the US, the expert estimate had to have used a lower probability of breakdown.

Why could that be?

If the average life of a transmission is 7 years, but if everybody has a brand new car, there will be very few transmissions being broken in a given year. So the probability of the transmission breaking must increase with time.

Excellent, in fact, we have found the following:



From this graph, we see that if the average age of the US car pool is relatively low, a lower percent of transmissions actually break every year.

So what would be the key take-away you would want to communicate to your client from this analysis.

Based on this analysis, I estimate the total remanufactured transmission market to be about $2.5 billion in value or about 3 million parts in volume. I have also found that a significant portion (probably somewhere between 75%-90%) of broken transmissions are not replaced with remanufactured ones. This has serious implications for my client. He might motivate customers to buy remanufactured transmissions instead of new ones by increasing perceived quality through advertising, trade shows, demonstrations or other means. He could also try targeting the lower end of the market - customers who are currently choosing to scrap their car instead of replacing the broken part. These people are price-sensitive, so he will have to offer them lower prices. Finally, I have found that the key driver of this analysis is the age of the vehicle. The age of cars in the total US car pool will impact the demand for remanufactured transmissions. If the portion of older cars increases, the remanufactured transmission market will experience high growth.

BCG 1st Round- 1

Case scope provided by the interviewer

Our client is a manufacturer of garden tools. They produce weedmasters, and they are market leaders. Their revenue has been dropping since 2000, 3-5% per year.

Additional information provided after relevant questions

- I learned that WalMart has introduced the same product with itsprivate label and selling it for a discount in its stores.
- Our client's FC is $30, where the price of WalMart's product is $30
- If we increase our price by 20% the demand for our product drops by 10%.

Example - How to crack the case?

We have to understand what has changed for 3 years. I would like to look at the Market (growth/shares/competitors/new entry to the market)
Customers (segments, change, needs)

In order to look at the change in profitability if we decrease our price, I asked if we had any price elasticity info. After the interviewer asked me to define price elasticity, she told me that if we increase our price by 20% the demand for our product drops by 10%. When I got this information, I said that instead of decreasing our price we should even increase our price in order to increase our profits.

Here, the interviewer asked me to calculate how the profit will change with a price of $40 and $60. After calculating and proving that the profit will increase, I took time for my final recommendation.

Recommendation:
Our client should not match the price of Wal-Mart, instead they should increase the price of their product. In the long-term our client needs to revise their marketing strategy and start to differentiate itself with high quality. The product should be sold in specialty stores or places like Home Depot and the product has to have special merchandising to emphasize its quality. The client needs to create brand equity through promotions and marketing.

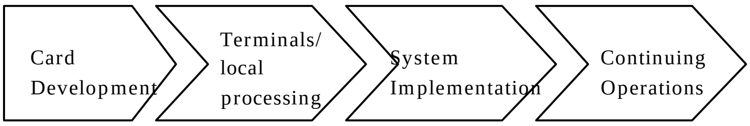
BCG 1st Round- 2

Case scope provided by the interviewer

Our client is a diversified electronics manufacturer and would like to enter the smart card industry. Smart cards are similar to a credit card in terms of application but instead of requiring a telephone call to gain bank approval, the chip is embedded with information that transmits electronically for approval. Where in the value chain should our client enter this business?

Conversational example - How to crack the case?

First I need to determine what the value chain looks like. It would start with the manufacture of the chip. Then the card readers would need to be manufactured. A service team is needed to install the initial network set-up. And finally a maintenance team will need to be available to service the readers. It would look something like this:



That's correct.

I'd like to look closer at each to gain a better understanding. Are all cards the same in terms of development?
No. There are four operating system standards each with equal market share.

How many companies are in the business of terminal processing?
It's a fragmented industry.

What kinds of companies are involved in system implementation?
These are typical IT Consulting firms and many are a part of larger firms that are in other businesses as well.

And how are the continuing operations different?
Continuing Operations is maintenance of the terminals. It is low-skilled, similar to ATM machine maintenance.

Now I'd like to explore the profitability in each part.
Good. What would you like to know?

What are the profit margins for each piece of the value chain?
Card Development: 15%
Terminals/local Processing: 10%
System Implementation: 20%
Continuing Operations: 5%

And what does the revenue distribution look like?
Card Development: 25%
Terminals/local Processing: 25%
System Implementation: 20%
Continuing Operations: 30%

If I took $100 in revenues and distributed it accordingly and then applied the proper profit margins, it would show that the greatest profits then reside at the System Implementation and then Card Development.

	Card Development	Terminals/ local Processing	System Implementation	Continuing Operations
Revenue %	$25	$25	$20	$30
Profit Margin	15%	10%	20%	5%
Profit Total	3.75	2.5	4.00	1.5

Since both of these businesses are similarly attractive from a profitability standpoint, I'd like to take a closer look at them. Do any of our current businesses manufacture chips (that could be used in card development) or provide System Implementation?
We manufacture electronics and have a small maintenance operation.

There may have potential synergies in manufacturing.
Let's say that our client can't garner any significant synergies from any piece of the value chain. What else would you consider?

The real power in this value chain is at the System Implementation level. The customer will only deal with the System Implementation, not with the card development and terminal processing.

There must be a different level of risk for each part. We already know that there are four different standards in card development operating systems. Can the System Implementation technicians service all four standards?
Yes.

Since there is no indication of which standard may become the dominant standard that would be a more risky venture. Additionally, once a standard is determined the chip used in card development will become a commodity product. Service is less risky since the service technicians can service are four standards. It is also easier to differentiate service and charge a premium price.

Why do you think are there more stripe cards in US and chip cards in Europe?

In Europe the telecommunication is newly deregulated, and still very expensive, whereas in US long-distance calls are very cheap. So in Europe, it makes more sense to store data in the chip rather than making phone calls each time to the server when there is a transaction.

Another reason that the interviewer told me, but not expected me to guess is: there is a French chip company, and EU is supporting them.

BCG 1st Round- 3

Case scope provided by the interviewer

The client is a Finnish coffee producer. Assume the case is taking place in the first half of 1990s. The client has been facing deteriorating profits. They want to know why and what to do about it.

Interview Hot Tips: This is a broad profitability case. Analyze the market first by asking about growth patterns, competitive landscape, different products and customers. Ask for any potential entrants and MS changes. Then proceed to inquire about the way the client operates in this market and analyze the financials. Customers and their preferences are also worth analyzing.

Conversational example - How to crack the case?

(After discussing the framework with the interviewer…) Let me start by asking about the market first. How is the growth in the market? How are the market shares of different players?
The market is flat and grows only with the GDP. It is quite a concentrated market. The client has 50% MS. The next player has 20%. The rest of the market is fragmented among a large number of players.

Is coffee the only product in the market?
Yes. It is pretty much a commodity in Finland.

Good. Let's proceed to our analysis of the company. How do our financials look like? I would like to start by an understanding the Price and Quantity trends of our company?
Quantity is the same for many years. Price is very slightly up.

How about the cost side? Have we seen any increase in the coffee bean prices or direct labor?
No, none.

How about the fixed cost side? Depreciation and SGA?
SGA portion is up.

Let's expand on that. SGA consists of general expenses, selling, advertising, promotion and admin costs. Which ones are up?
All selling and marketing related costs are up. Why do you think?

It looks like a peaceful market to me. All of a sudden something must have happened. My initial hypothesis is that there is a new entrant. Another possibility is that our client decided to steal market share from the competitors or competitors decided to steal from us. However the absence of a drop in prices signals otherwise. Also consumer preferences might be changing.

Your first hypothesis is true. Nestle entered the Finnish market. Why do you think they entered then and not earlier?

(Keep in mind that such a question never comes out of the blue. He was testing my ability to establish the bridge between what I have heard so far and what is to come next). You said the case took place in early 1990's. I would assume that Finland joined EU and the trade barriers came down.

That is correct. Nestle launched instant coffee in Finland. The local Finnish coffee is different from instant. Actually the second biggest player with 20% MS is Nestle. They went up to 20% in 3 years. Now tell me. Why do you think the coffee market in Finland did not grow and rather Nestle stole away market share from local competitors?

The first thing that comes to my mind is the possibility that Finnish coffee market is very saturated and the consumption is actually at peak. To assess that I would compare per capita consumption of Finland with another "coffee-intensive" country like the US. Does this sound reasonable?

Actually what you said is entirely correct. Finland is the country where the per capita coffee consumption is the highest.
Why don't you summarize me your findings and state your recommendations?

(As always, take the full minute to jot down your recommendations. Associated with each one, state the benefit, the cost and the risk (if any) of implementation).

(After a brief summary of findings) What the client can do to tackle the prevalent problem is two folds: In the internal market, it can choose to penetrate the current market by introducing different flavors and aromas of the Finnish coffee. Whether this is feasible and whether the people will buy it is of question, of course, but a detailed market research can indicate some hints.

The problem at hand is more like the protection of the local market rather than growth. Nestle is a very strong brand with deep pockets. It also has already established a beachhead in Finland. At this point for the local producers it is a survival game. Therefore, some tacit collusion with the other players (which adds up to 80% MS) and starting a campaign against instant coffee and promoting the local coffee as something like the “Finnish Inheritance” could work against Nestle. However, somebody can always defect.

Another option can be to diversify into new businesses like opening up coffee shops like Starbucks. The risk of that is the lack of necessary expertise and know-how of the business.

Of course, the most natural option will be to find other markets to sell. Another country with a similar taste, or a place where Finnish coffee can be promoted as an “exotic” or original taste could be a good export target. This will also give Nestle that the client is willing to fight the battle also outside Finland.

BCG 1st Round- 4

Case scope provided by the interviewer

The client is a manufacturer of a commodity product and produces and sells only in Korea. It is the Korean arm of a US comoany. I will not tell you what the product is for the time being. For the first time in its history it was able to increase its market share but it incurred a loss. We have been called in to understand what is happening and what could be done.

Interview Hot Tips: Another broad profitability case. Analyze the market first by asking about growth patterns, competitive landscape, and customers. Ask for MS changes and typical profit margins. Then proceed to inquire about the way the client operates in this market and analyze the financials.

Conversational example - How to crack the case?

(After discussing the framework with the interviewer…) Let me start by asking about the market first. How is the growth in the market? How are the market shares of different players?

Let me ask you a question: can you tell me the profitability equation?

Profit = Revenues – costs. Namely pQ-VCQ-FC.

Now what if I tell you that FC, VC/unit and price are the same as last year?

Then the loss is entirely coming from the loss in quantity. This, in return, means that the shrink in the market size is greater than our loss in quantity.

That is correct. Coming back to the questions you had asked previously, the client has 40% MS and the rest of the market is divided among 4 players with 15 to 25% MS each.

How does this number compare to the “good old days”?

In the past there were 12 players in the market.

This means there has been a consolidation in the industry and some players have exited the industry. Before going further I have a question: You said the client was a US company. Does it incur any losses due to depreciation of Korean money unit against the US Dollar?

Excellent question. Assume no appreciation or depreciation.

Then let me understand the way this company operates. I want to have a look at the value chain.

Okay. Time to tell you the commodity. It is audiocassette. Simplifying things, assume that client produces and sells to recording studios and they sell to consumers. Now, Why don't you immediately tell me your initial recommendations at this stage?

Most probably, due to the obsolescence of the cassettes and the switchover to CD's the industry has a tough time in selling what it produces. At this point, the immediate solution that comes to my mind is to make the necessary investments to keep up with this switch.

Assume that the client had poured a lot of money to cassette business and they want to skim it as much as it can. Also assume that the local market has only 4-5 years before becoming "extinct".

Then, possible options are to look for a country where there is still demand for cassettes, Such places could be China, India or any other developing or underdeveloped country, which is unlikely to go for CD's in the short run. Another solution might be to go for a similar line of business like data tapes for data backup or micro cassettes for answering machines or voice recorders. Given the short life expectancy of the market, I would not go for buying out a competitor.

Good. Assume that we plan to go to China. What would be the things to look at.

(Hint: the case is taking a different shape now. Request another minute and come up with framework number two)

(After one minute of thinking and developing an action plan…) I would first question whether there is enough room in this market for one more competitor. Next I would question the profitability in this market to see whether I can really make money in China. Finally, I would look at the competitive landscape in the market and decide on the mode of entry, which will be made harder through tariffs, logistics etc in China.

Assume that there is enough room in the market. Also assume that the Chinese market is 500 times as big as the Korean market and that an average Chinese competitor is 50 times as big as our client. Can you tell me how many such competitors are there in the Chinese market?

The Korean market is 2.5 times as big as the client. Therefore the Chinese market is 500 x 2.5 = 1250 times as big as the client. Since average competitor is 50 times as big, there are 1250/50 = 25 such comopetitors.

Good. What do we do next?

My next questions will be, whether the competitors are existent for a long time, whether they have deep pockets to crush us if we enter alone and whether we have any competitive advantage over them in terms of cost or quality. I am trying to get an understanding of how hard it will be to acquire customers.

They are existent for a long time, have deep pockets and making big money. Quality is not an advantage. However, we have a substantial cost advantage over them.

Is it possible to quantify this? How much do we produce for compared to them. Also I would like to learn the prices in Korean and Chinese markets.

Prices are the same: $1.40. They produce for 1.20 and we produce for $1.13. What are the possible additional issues that you need to consider in China?

In spite of the fact that it looks like a very attractive market, whether the competitors will let us live is questionable. Besides, we might have no knowledge of the local market. Also, there will be tariffs and barriers. Finally logistic conditions are very hard to tackle in China. Does this $1.13 include everything?

Yes, it is the marginal cost and everything is included. Now, how would you want to enter this market?

There are two possible ways: a hostile entry, through which we will go by ourselves and drop the prices to acquire customers, for there is no other way to differentiate. However, this will be costly and given our bad financial situation we will be vulnerable to retaliation. The other alternative is to use our cost advantage and become a supplier to one of the competitors.

What's in it for them? Why should they do it?

We will be producing a cheap source of production to them. However, to quantify the true benefit I need to learn their and our capacity utilization.

Currently we are running at 20% whereas they are running at 150%.

Great news. This means that we are on the economies of scale side of the long run average cost curve, whereas they are on the diseconomies of scale part. Every additional unit we can add to our capacity will decrease our cost whereas every additional unit they can add to their capacity will decrease their costs, too. It is a win-win situation. However the cost drop for them will not be same for them due to their sheer size.

Give me a price then.

I would start negotiating from $1.19 until a point where we can recover our cost of capital plus marginal costs.

Good. Thank you.

BCG 1st Round- 5

Case scope provided by the interviewer

A US global player which manufacturers machines that make prints in cans (like valid dates) is losing market share. Why? What can be done?

Additional information provided after relevant questions

	Players				Mkt Growth Rate / year
	client	**2**	**3**	**4**	
US	35%	25%	35%	5%	5%
Europe	40%	30%	25%	5%	3%
Asia	2%	10%	25%	63%	18%

- Market shares have the same historical pattern with the exception that player 4 is growing faster in Asia.
- (When I made the question about growth rate of the different markets, the recruiter gave me the info and asked, why do you think it is growing so fast in Asia? I asked: 1- GDPs of many countries are increasing faster than in the rest of the world, as well as per-capita income, and can consumption grows as a result; 2- multinational players may be entering Asian market and developing this untapped market; 3 - new potential regulation may have obliged can manufactures to print stuff on their cans).
- Profitability in Asia is even higher than in US and Europe.

Example - How to crack the case?

So, Asia is an attractive market which is growing very fast we our client is not well positioned. What can we do about it?

- client has US employees as the management of the local business in Asia.
- client's product requires that end-user (local can manufactures) have some minimum requirements that are very expensive to acquire. This limits the size of the market.
- client does not spend time and does not focus on understanding end-users local needs.
- client has the resources and skills to succeed in Asia as well.

I first focused on understanding the company's products and clients. I wanted to know who are the clients and their locations (clients mix by type of order, by industry, by geography, etc). Latin America and Africa were not considered as part of the overall market for matters of simplicity. Once knowing the locations in USA, Europe and Asia I wanted to know the growth rate of each region. I was already guessing that the client was not "playing" well in the fastest growing region. That was proved when I got the data. Therefore, the reason why mkt share was decreasing was explained.
Now I needed to provide recommendations. I wanted to understand why the client was not growing in Asia. Understanding that (data above) I wanted to know whether the success in US and Europe could be replicable to Asia (resources, skills). Once it could, I recommended to launch a appropriate product / service to Asia and hire local management to leverage the company's market share in the region.

BCG 1st Round- 6

Case scope provided by the interviewer

Your client manufactures“out-of-the shelf” steel laminates. He is analyzing the opportunity to acquire a player that produces “custom-made” steel laminates. Should he perform the acquisition?

Additional information provided after relevant questions

- NPV = $80M (for the stand-alone business, not considering synergies with acquirer).
- Seller was asking $100M for the company.
- Raw material was the same and purchases costs could be reduced by 10% if companies bought together. Total raw material cost (combined) was $90M.
- Manufacturing could be improved on acquirer by using best practices of target. Another 5% could be reached on $70M.
- 50% of clients that buy “custom-made” steel laminates also buy “out-of-the shelf” and vice-versa. Revenue of acquirer was $120M and of target was $90M.
- Margin was 10% for custom-made and 5% for out-of-the shelf laminates.

Example - How to crack the case?

I started showing that if the NPV were higher than the deal price, than the company should move forward. He said the NPV was $80M and the price asked by the company was $100M. Then, I said, now what we have to understand is whether the synergies are greater than $20M.
I draw the value chain (from purchase to sales) and indentified where synergies could be reached. Then I calculated the synergies and they would be lower than $20M.
Initially revenue synergies appeared to be great because of complementarity of products and client base, but there was one problem: one company was on the east coast and the other on the west coast. The products had high transportation cost, so clients were all located close to the suppliers. The products, although with a competitive price locally, would be priced higher than competition when shipped to the other coast. Because it was a commodity, price was king! Without the revenues' synergy the deal was not worthy therefore no acquisition should be performed.

BCG 1st Round- 7

Case scope provided by the interviewer

A cereal manufacturer has two alternatives to choose from to grow their current business. Option 1 is an adult cereal. The manufacturer has developed a proprietary technology, which will let them claim the cereal has substantial health benefits. Option 2 is a kid's cereal. The manufacturer acquired the rights to market this cereal as a "Harry Potter" cereal. The CEO would like to know which option should the company choose?

Additional information provided after relevant questions

- Industry growing at 5% per year and steady
- Product mix: 40% children, 20% adult, 40% various breakfast products
- Adult cereal:
  - Price = $1.50
  - VC = $1.00
  - FC = $10M
  - Total pounds in market 700 million
- Kids cereal:
  - Price = $1.60
  - VC = $1.00
  - FC = $12M
  - Total pounds in market 1.4 billion

Example - How to crack the case?

Initially I asked questions to get a handle on the company's current product mix and industry/competition information. The bottom line question is which option allows the company to make the most profit – therefore, following the structure of a profit case while solving for break-even volume yields the best solution.

- Adult cereal:
  - Contribution Margin = $.50
  - BE Volume = 20M pounds
- Kids cereal:
  - Contribution Margin = $.60
  - BE Volume = 20M pounds
- Achieving 20M pounds in the kids market will be much easier (lower percentage of total market) than the adult market. Assess the risk of how popular Harry Potter

will be to children in the future– the interviewer will let you conjecture as you see fit.

BCG 2nd Round- 1

Case scope provided by the interviewer

Your client is a league such as the NFL or NHL. The league has a business that makes money by licensing out team logos to manufacturers of clothing and other sportswear such as Nike and Adidas. Although revenues have been rising they are not seeing an increase in profitability. How should they fix this problem and make sure that it does not reoccur?

Hot tip: this is a basic profitability question; therefore, the appropriate framework and thinking should be employed

Conversational example - How to crack the case?

Before I get started I'd like to know a little bit more about the company, which is, how they produce revenues and how their operations are set-up.
The league receives a percentage of sales of the products that use its logos; they are not involved in any other revenue generating activity. They have an office but do not get involved in any of the retail portion of the business; they only deal with big companies like Nike and Adidas.

At this point I laid out the profitability framework and reviewed it for my interviewer. I used the three C's (competitors, company and customers) indicating that I was looking at them as background information; however, I emphasized that the source of the problem would be explored using Profits = Revenue – Costs.

After a brief overview of the three C's and the market we explored the profitability equation:

Year	1991	1992
Sales	91	100
Costs	42.5	50
Profits	48.5	50

These were the only two years he wanted analyzed.
(Interviewer: Feel free to give partial information and then ask the interviewee to do some math.)

Profits can be given as just straight profits or sometimes they are presented as a percentage of sales. In this case what is changing more quickly: profits or return on sales? One of these calculations is very easy to calculate.

Return on sales is changing more rapidly.
That is correct.

I'd like to look at the reason why costs are increasing, therefore, I would like to take a closer look at the cost structure of the organization.

What would you like to know?

I would like to know what the costs are and how those costs have changed between these two years. I would like to examine both the fixed and the variable costs.
What do you believe are some of the overall costs of running this business?

Fixed costs: SG&A (including advertising and promotion), Depreciation
Variable costs: Labour, Overhead like rent, heat, phone etc., Cost of sales

Ask the interviewee to describe the components of SG&A prior to telling him/her the following: Advertising and promotional spending has increased, right now the firm does trade shows and catalogs forthe companies they sell to.

What other costs could have accounted for the total increase in costs? (I got stuck here until I realized it was the legal or patenting costs)

	1991	1992
Advertising and promotion	11.25	15
Legal	11.25	15

(The interviewer lumped all other costs into an 'overhead' section which had costs of 20)

The interviewee should show all calculations and account for the increase in costs and relate that to the flat profitability.

Student tip:

Be sure to include reasons why and how the company can reduce or contain costs; I threw in a few revenue generating ideas as my 'additional suggestions' section at the end.

BCG 2nd Round- 2

Case scope provided by the interviewer

The client is a cardboard box manufacturer. Recently a competitor announced its intention to enter the market by the new PVC boxes and that they will charge 80% of whatever the client charges. Both boxes are/will be sold to CPG manufacturers like P&G, Unilever etc. The client came to us and asked us to assess whether this threat is credible. Might there be any hidden costs to make it less attractive

Interview Hot Tips: A completely out of the box case about, ironically, boxes. The last sentence is the key. Look for the hidden costs for the customers and the manufacturers only. Interviewer should make sure that interviewee is focusing on these items at all times. There are no numbers in this case. It is totally qualitative.

(After examining a whole lot of irrelevant issues like financial risks born on the manufacturer by the “80%” commitment, anti-dumping, predatory pricing etc. etc.)... Now I want to compare the processes that the PVC box goes through with those of the cardboard box.

Good. Let's do it.

Let us start from the time when the boxes arrive first at the inventory of, say, P&G. I would think that the PVC boxes have a fixed form, whereas the cardboard boxes come in flat layers and can be bent to whatever size CPG manufacturer desires.

Good. What does this mean?

This means that the storage cost of a PVC box is considerably higher than the cardboard box. Besides the rigid form of PVC boxes makes me think that there cannot be an available size PVC box for every need.

What does this imply?

This implies inefficiencies in the box filling process. Since P&G cannot fill the boxes fully, it is paying extra transportation costs because it cannot utilize the trucks fully. There are also additional costs involved because of added complexity to hold so many box sizes in inventory.

What else?

Associated with this, comes the additional cost of filling material to tackle with the breakage problem (in case of breakable items) due to movement inside the box. Then comes the situation in the trucks. Since the boxes are not of the right size and, probably, will mostly be bigger than necessary, we'll have extra costs due to inefficient truck-loading for the second time.

Makes sense. What else?

On the other hand, since PVC is more durable than cardboard, it will be less affected by, humidity, breakage if it falls from the cranes etc. PVC will have clear advantage there.

What else?

The storage problems we have talked about are also valid for the retailer. When they open up the boxes, they will need a place to store all those boxes of various sizes. By the way, what happens to these boxes in the end?

They will be picked up by the manufacturer and taken to a pool to be resold to CPG manufacturers. Assume, on average they can be used 20 times before recycling, whereas cardboard boxes are mostly disposed.

So the manufacturer of the boxes will be, theoretically making a lot of money, if it can achieve the customer base. There is one thing I might have missed on the way, though. I never talked about the production process at CPG manufacturer. Now they have a production line designed for cardboard boxes. They will have to make the necessary investments to convert to PVC.

Good. Can you think of something else?

Yes. There is worse: some customers will adopt it and some not. Therefore CPG manufacturers will have to run two production lines. That means extra complexity and cost.

Great. Why don't you summarize and come up with a recommendation?

(After a brief summary…) Despite the fact that we have not quantified anything, it is obvious that the extra costs offset the 20% price differential. Therefore the threat is not credible.

BCG 2nd Round- 3

Case scope provided by the interviewer

I received a call yesterday from a potential client. They are a retail chain similar to Neiman Marcus. They would like to increase revenue. You can assume that all products have a 75% margin.

Example - How to crack the case?

- I would like to look at three areas. I would like to investigate increasing sales of existing products in existing stores, increasing sales through new products in existing stores and increasing sales through the expanding the retailer's "reach".
- To investigate the possibility of increasing sales of existing products in the current stores I would be to look at the market or customers, the competition in this segment and the company itself. (I just got thecall yesterday and I don't know anything about the customers, company, or competition. Who do you think the customers are?)
- Neiman Marcus is a very exclusive an expensive retailer. I expect the customers are not sensitive to price, that they demand the m ost prestigious brand. I expect that they are very affluent. I expect that they are likely to be limited in the amount of time spent shopping. (Sounds reasonable)
- My expectation is the competition is muted for high-end retailing. I would expect that they focus on service instead of price. I have heard that Neiman Marcus allowed a customer to return four tires to one of their stores even though Neiman Marcus doesn't sell tires.
- The company may be able to increase revenues of existing products in existing stores by drawing more customers into the store. Neiman Marcus keeps a record of customers and their purchases. Their sales associates also give out business cards and work to develop personal relationships with customers. A sales associate could call or send a personal letter with the message "you purchased a solid black Armani suit last month. We also have a solid navy that might compliment your wardrobe.
- The company might also be able to entice customers into the store by sending a special invitation for an exclusive meeting with one of the premier apparel designers carried by the company.
- Now I would like to look at opportunities to increase sales by increasing the products and types of products sold. With all products carrying a 75% margin

and a price insensitive customer base I would want to carry more high-ticket items. I have seen Neiman Marcus carrying $150K fur coats, and automobiles. That is a possibility for increasing the revenues and profit.

BCG 2nd Round- 6

Case scope provided by the interviewer

Our client is a fruit juice producer and seller. They have been experiencing an increase in revenues but the profits are flat. What might be the reasons behind this situation and what can we do?

Additional information provided after relevant questions

- Our client has several production lines.
- The production process consists of several steps like mixing the fruit concentrates, adding other supplements and bottling.
- We had to hire more labor in order to satisfy the increasing demand.
- Our raw material cost/unit did not change in the recent years.
- Our transportation cost/unit did not change in the recent years.
- We have an increase in the processing cost/unit
- Also we have an increase in the marketing costs.

Example - How to crack the case?

First of all, you should ask a few clarifying questions. You have to understand the client's production model. While asking questions, you have to show your field and production experience. You have to mention about the production lines, you can draw sketches etc. Always be curious about the possible cost buckets. Try to visualize the production in your head and ask questions to fill in the gaps.

The big picture of the case is that, although the revenues are increasing, profits are flat. This means costs are also increasing. So after understanding the production model and the products, directly ask questions about the costs. Create a cost framework and show all the cost buckets. Go over all of them. You will see that there is an increase in process costs per unit. You should put a question mark near this bucket, showing the interviewer that you will dive more deeply on this issue soon. Cover the remaining cost buckets. When you return to the processing cost issue, use all the information and sketches you come up with in order to identify the problem. You will see that, the tanks have to be cleaned between the productions of every different fruit juice type. So there is a fruit juice and time loss in every cleaning process.

At this point you can make a summary. In 8-10 sentences talk about the problem, products / production and tell the main problem issue you have found out. Now you can start your recommendations. Take some time and think about the recommendations. Always ask the "so what?" question to yourself.

Because the main problem is the "setup time costs", you have to give recommendations to decrease this cost. Recommend attaining some production lines only for high volume juice types, so you will eliminate setup times. The juices should be distributed among the production lines according to the "minimum setup time" criteria. So some low volume juice types can be out sourced. The client has to make a cost/benefit analysis in order to decide the fruit juices to be outsourced. Talk about the risks of outsourcing: less quality control, less control in timing etc.

DiamondCluster International – 2nd Round

The client is a company that is operating in the roofing industry. It is a 70 year old company. Currently the sales of the products (wood, nails, roof tiles, etc.) are down 5%. However, the services business is growing rapidly. Services include consulting on roofing projects, providing design help, etc. The company wants to stWhat should we tell the company to do?

✓ Interview Hot Tips: This case is certainly set up for a profitability framework. However, the interviewer just wanted to talk on a macro level.

Conversational example - How to crack the case?

(After discussing the framework with the interviewer…) I would like to first look at the current situation with this company. Is it safe to assume the products business fluctuates with housing starts and the general economy and thus this problem is endemic to the industry as opposed to a problem experienced solely by our company?
Yes. That is safe to assume.

Okay. Given that, we probably don't want to make a rash judgment as to the viability of the products business. We would probably want to make the company more immune to the fluctuations, but probably don't want to exit the area.
True.

I want to understand now what percentage of our business we should orient towards products and what percentage we should allocate to services. What are the margins like in both of these businesses?
Well, the products side of the business is a high margin –close to 70%. However, the services business is also a high margin business. The key to the services business is covering the most area in an efficient manner so that the COGS is decreased.

Then let me ask about the services business. How do we currently sell our services? Do we just piggyback off product sales or is a totally separate sales effort?
Well, many of our services are sold at the product point of sale – in other words, when people come in to purchase roofing products, we also inform them of our services business. However, the company just recently started a sales team to help grow the services business. We are trying to figure out how to set up the sales team to best sell these services. How would you organize this?

Well, first I would try and assess all the possible alternatives. We could set up a telesales operation, set up regional person-to-person sales network, or we could do a combination of the two. We could also look at partnering with another company who sells to the same target group of customers and have them sell our service business as a value added proposition. Of course, with this effort we would need to figure out some sort of revenue sharing scheme. I am sure there are other approaches to could be used to effectively

Okay. While I think the telesales is an interesting proposition I don't think we can effectively sell such services through that channel. We really need to visit our clients and create relationships.

Great. So let's look at how to organize the regional sales network. In order to cover the greatest area, we would need offices in major cities, where lots of construction projects happen. Then those sales people could head out to the suburban areas where more business could be generated.

Sounds good. We originally started in this fashion- basically selling to existing business verticals. How would you suggest we sell our services to new verticals?

We need to really have an efficient sales force that targets customers. We should certainly leverage best-in-class sales operations from other firms that are selling similar services – like consulting firms. We need to use comparables to start fashioning our sales force. I also think partnering with firms who have relationships in the verticals we want to enter would be helpful.

Okay. Of course there is lots more to discuss with respect to the sales team. But I want to shift gears and talk about synergies between services and sales. Tell me what you see as the synergies between the products and services area.

These business segments are very complementary. We can use product customers as the core base for our services business and then expand from there – using this initial base as a way to achieve a good reputation and to generate referral sales. We must leverage our relationships with industry, as their products provider, to try and increase business and to get our name out.

Great. Now, how would you set up the management of this company? Would the services business be totally separate from the products business or would the management run both from a central office. Tell me what you think and the reasons behind your thought process.

I think we need to have both the services and products business under the same management umbrella because of the synergies I identified before. However, the two businesses should be seen as separate profit centers because of the fact that, operationally, they are vastly different businesses.

Well – I just wanted to see your thought process on a macro level about some of the issues confronting this company. Thanks for your efforts.

DiamondCluster 2nd Round- 1

Case scope provided by the interviewer

Our client is a leading ISP and has recently seen decline of customer base and resulting loss of profits. They would like you to investigate this problem and come up with recommendations.

First, I would like to know more about the company. What are the products of the company besides Internet Service?
Well, there are other businesses in the corporation. They are in media and cable business.

Do they provide both narrow band and broadband service?
Yes, they have recently started the broadband service

What about the competitors? Who are our competitors?
A couple of big competitors and many small fragmented service providers

We are losing customers, what about prices? Are they holding?
Prices haven't dropped significantly over the last couple of years

I am guessing that many of our customers are moving to broadband.
Yes, that is correct.

Do we know if we are cannibalizing our narrow band business?
Yes, many of our customers are moving from narrow band to broadband. But we are also losing many to other broadband providers

I see. Now I would like to focus on our customers. Who is our typical customer? Age group? Sex?
Well, there are a lot of young customers. There has been growth in that segment recently

What are the value drivers for these customers?
What do you think?

Probably music programs, maybe fashion oriented content.

Ok

I think we have cross-selling opportunities with our narrow band customers.

What are the challenges in doing that? Tell me about the organizational challenges in bringing about this change

Well, we'll have to look at various functions within the organization in the context of an organizational push toward broadband. One approach might be creating a high level task force, which will assign a broadband representative in each functional group. I think it will be important to make the transition to broadband without creating a threatening environment for existing employees. Toward this end, maybe training sessions can be arranged for employees

What about the cross-selling? How do you make that happen?

We could use some of the modern available technologies here. For example, the representative handling customers in a call center has the ability to see customer data on the screen as he or she speaks to the customer. They can make a pitch to the customer for broadband service and sell different features of the service based on the demographic group of the customer

Can you please wrap it up now?

I would say that strategically our client needs to take the following steps in order to regain profits:

- Focus on the broadband business and reorient the organization toward broadband service
- Leverage the content from the media business for the broadband offering. We'll have advantage over our competitions in this area.
- Especially focus on the youth segment by providing music, sports and entertainment
- Cross-sell aggressively to existing customers
- Follow a premium content and pricing strategy for certain market segments

DiamondCluster 1ST Round- 1

Case scope provided by the interviewer

How many classes are going on in the University of Michigan at 11:00 a.m. on a Tuesday?

Example - How to crack the case?

This was a market sizing case.

1. Estimated the total student body of Michigan to be 40,000
2. Said 35,000 full time day time students out of the 40,000
3. Divided the 35,000 into sections of 60, so got 600 sections
4. Said that each section is taking at least 2 classes per day, so there are 1200 classes per day
5. If classes begin at 8:30 and end at 5:00, than we 5 possible classes sessions going on during the day.
6. So 1200 classes in 5 sessions, gives us approximately **240** classes at 11:00 a.m.

Very good, you can given me one approach to come up with this number, can you come up with another approach?

We can also quickly look at the total number of classes running in the business school at 11:00 a.m. and having known that, scale that number to the entire university.
Any other approach? What about a supply side approach rather than a demand based approach?

We could also estimate the size of the faculty, than based on an estimate of the teaching hours for each faculty member, come up with an estimate for total number of classes that can be offered on any given week day.
Great, thank you very much.

DiamondCluster 1st Round- 1

Case scope provided by the interviewer

My friend is a consultant and he travels a lot. He uses the limousine service often when he travels and he has identified this as a good business opportunity. He wants you to tell him how many limousines he needs to have in his fleet in order to make money in this business?

Conversational example - How to crack the case?

In order to approach this case, I will first look at all the costs incurred and revenues generated in this business. Than, I will apply the break-even equation for finding the limousines required to operate the business without a loss.

What is the cost of each limousine?
The cost to buy outright is too high. We will lease the limousines and that cost will be $2000 per month.

What other fixed costs will the business incur?
What do you think?

How about renting office space?
Yes, we will be paying $30,000 per year for office space.

How about salaries?
Let's say we will pay our drivers 20% of the fare for each ride.

Do we have an estimate for how much profit we make per ride?
Well, we charge $20 per ride on an average

Do we have an estimate of the number of rides per limousine per year?
What do you think?

Well, let's say we work 300 days a year, and an average of 12 hrs of work per day, and say one ride per hour. That makes it 3600 rides per year. So each limousine is bringing in $72,000 per year.

Now, I will like to know the variable costs incurred. What about gas? Any other costs? Maintenance? Insurance?
He gave me estimates for all of those costs.

Proposed solution:
At the end of all this, I had the total annual fixed costs and an estimate of the contribution margin of each limousine per year. To get the number of limousines, I simply applied the Break Equation, # of limousines = Total fixed costs / Contribution margin per limousine.